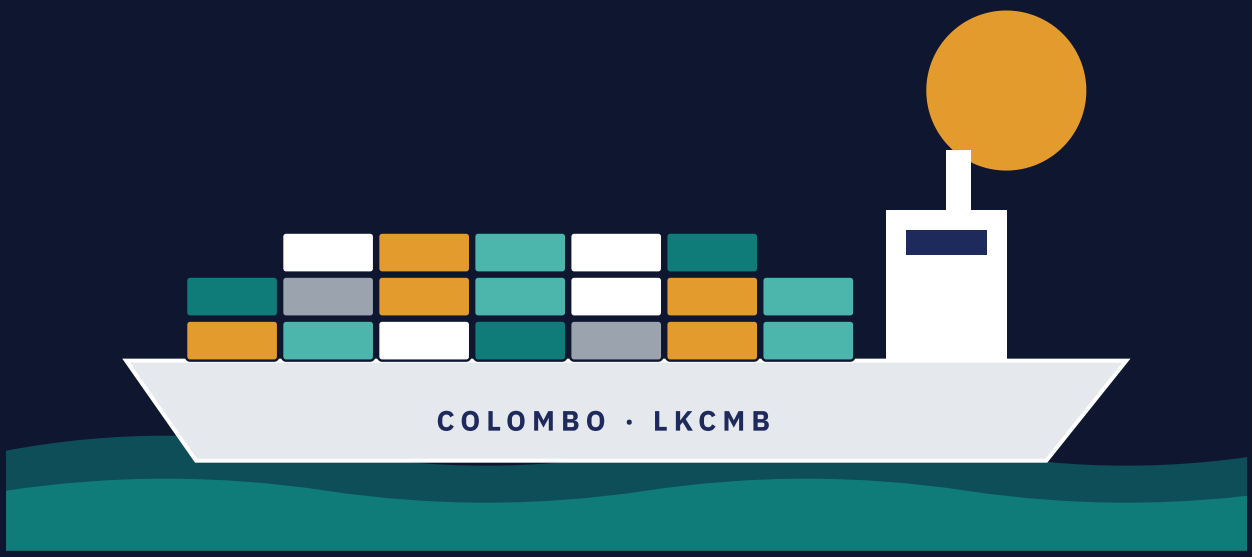
Ternary
SYSTEMS

Cargo, explained simply

A beginner's booklet on shipping, logistics, documents and customs, and a guide to using **Ternary Cargo**.

🚢 Sea · ✈️ Air · 🚛 Road · 📦 Containers · 📜 Documents · 🛂 Customs · 💰 Money



Prepared for **Roshan Gamage**, Founder, Ternary Systems
Colombo, Sri Lanka · 31 October 2026

Research. Invent. Deliver.

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How to use this booklet

Hello Roshan! This booklet explains cargo from the very beginning, as if to a curious 10-year-old, but with real facts you can rely on. Read it front to back once, then keep it as a reference.

Meet the boxes

> 💡 **Think of it like this**
> Everyday comparisons that make a new idea easy, like “a container is a giant lunchbox”.

> 🇱🇰 **A Sri Lankan example**
> Examples from our own country: Colombo port, Bandaranaike Airport, Sri Lanka Customs and Sri Lankan exporters of tea and clothes.

> ⭐ **Top tip**
> Practical advice that saves time and money.

> ⚠️ **Watch out!**
> Common mistakes and safety rules.

> **Remember!**
> - Every chapter ends with a **Remember!** box: the few things to keep in your head.

Meet our pretend companies

🍃 **Ceylon Leaf Teas**
A Colombo tea exporter selling black tea to Hamburg, Germany.
Sea FCL

👕 **Lanka Knit Apparel**
A Katunayake garment factory selling T-shirts to the UK.
Sea and air

🧭 **Ceylon Freight Lines**
The freight forwarder in the Ternary Cargo demo.
Forwarder

The big picture in one sentence

A seller packs goods, a forwarder books a carrier, customs check them out of one country and into another, and the buyer collects them, while the right papers and the right money move alongside.

Chapters 1 to 12 explain how cargo works. Chapter 13 shows you how to do it all in **Ternary Cargo**. Chapter 14 is the A–Z glossary and cheat sheets.

CHAPTER 1

What are cargo and trade?

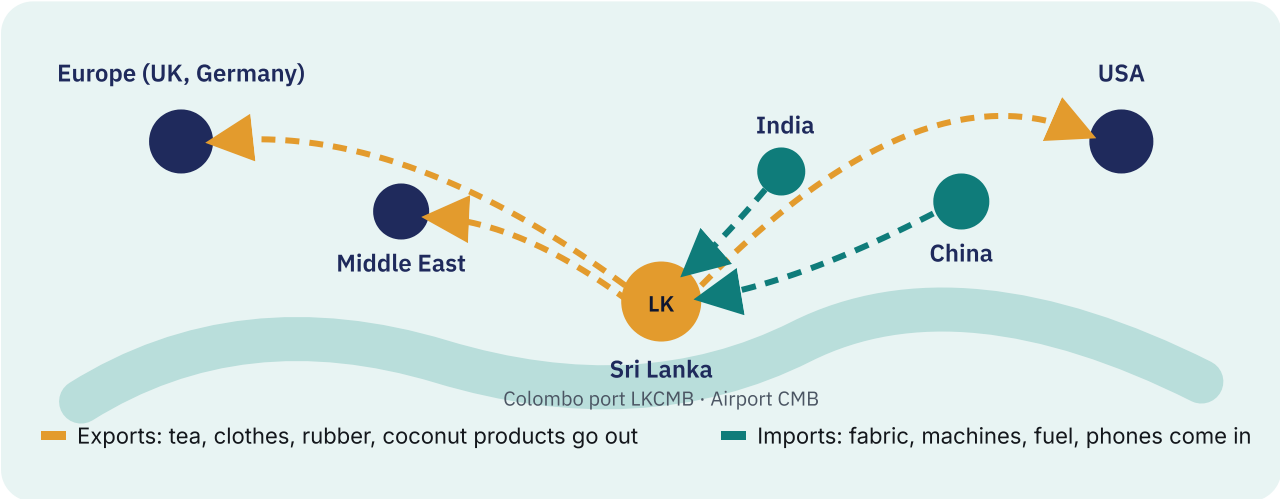
Cargo is simply *stuff that travels*. Trade is when people in different places buy and sell that stuff. Let's meet everyone who helps it move.

In this chapter you will learn: what cargo is, why countries trade, and the 13 main helpers (we call them *parties*) in every shipment.

Cargo is stuff on a journey

Imagine you bake 1,000 cupcakes in Kandy and your cousin wants to sell them in London. The cupcakes cannot walk. Someone has to pack them, carry them, check them at the border and hand them over. While they are travelling, those cupcakes are called **cargo** (you may also hear **freight** or **goods** – they mean nearly the same thing).

Trade means buying and selling. When a Sri Lankan company sells something to a buyer in another country, that is an **export** (it goes *out*). When a Sri Lankan company buys something from another country, it is an **import** (it comes *in*).



Sri Lanka trades with the whole world. Things we sell go out (exports); things we buy come in (imports).

> 💡 **Think of it like this**
>
> Trade is like swapping lunch at school. You have lots of tea; your friend has lots of apples. You swap, and you both get something you want. Countries do the same, only with money in between and with ships and planes instead of hands.

🇱🇰 **A Sri Lankan example**

Ceylon Leaf Teas (a pretend company we will follow in this booklet) packs black tea in Colombo and sells it to a shop chain in Hamburg, Germany. **Lanka Knit Apparel**, a pretend clothing factory in Katunayake, sews T-shirts for a brand in the UK and imports cotton fabric from China. Real Sri Lankan exporters work just like this.

Why do we need so many helpers?

Moving cargo between countries is like a relay race. No single runner does the whole race. Each helper carries the cargo, or its paperwork, for one part and then passes the baton. Here is the team.



Manufacturer

Makes the goods. Might be the same company as the exporter.

e.g. a garment factory in Katunayake



Exporter (seller / shipper)

Sells the goods and sends them out of the country. Often called the **shipper** on documents.

e.g. Ceylon Leaf Teas, Colombo



Importer (buyer / consignee)

Buys the goods and receives them. On documents the receiver is the **consignee**.

e.g. a tea shop chain in Hamburg



Freight forwarder

The trip organiser. Books ships or planes, trucks and paperwork for you, like a travel agent for boxes.

e.g. Ceylon Freight Lines (demo company)



Carrier / shipping line

Owns or runs the ships and containers. Issues the bill of lading.

e.g. Maersk, MSC, CMA CGM, ONE



Airline (air carrier)

Flies cargo, in the belly of passenger planes or on freighters.

e.g. SriLankan Airlines cargo at CMB



Customs broker (CHA)

A licensed expert who fills in and lodges customs forms for you. In Sri Lanka: a **Customs House Agent**.

Lodges the CUSDEC in ASYCUDA



Customs

The government's border guards for goods. They check what comes in and goes out and collect duty and taxes.

Sri Lanka Customs



Port / terminal

Where ships load and unload. Giant cranes lift containers on and off.

Port of Colombo (code LKCMB)



Warehouse / CFS

A big shed to store goods. A **CFS** (container freight station) packs and unpacks shared containers.

e.g. a CFS in Peliyagoda



Haulier (trucker)

Drives the truck that carries the container between factory, port and warehouse.

Colombo port to Katunayake



Bank

Moves the money safely, and can promise payment with a letter of credit.

Seller's and buyer's banks



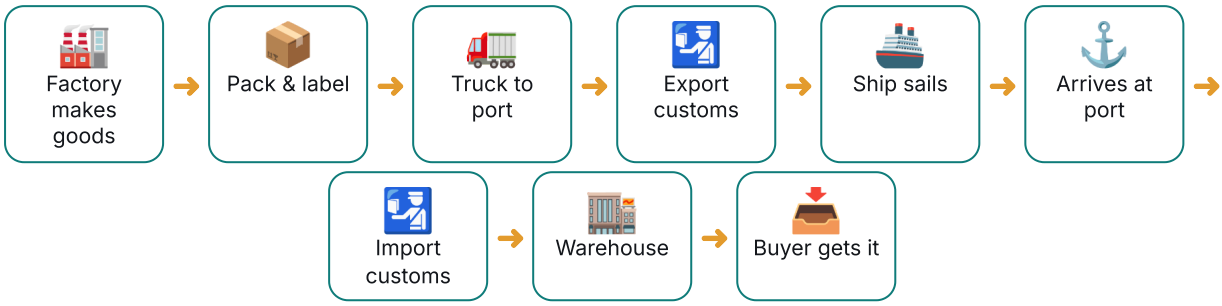
Insurer

Pays you back if your cargo is lost or damaged, if you bought cover.

Marine cargo insurance

There are a few more helpers you might hear about: **NVOCCs** (companies that sell space in containers without owning ships), **surveyors** (who inspect damaged goods), **inspection agencies** and the **port authority** (in Sri Lanka, the Sri Lanka Ports Authority, SLPA). And at the airport, **ground handlers** load and unload the planes.

How the relay race looks



> ⭐ **Top tip**
> On every shipment, ask two questions: **"Who has the goods right now?"** and **"Who has the paperwork?"**. Most problems happen when the goods and the papers are in different places.

Remember!

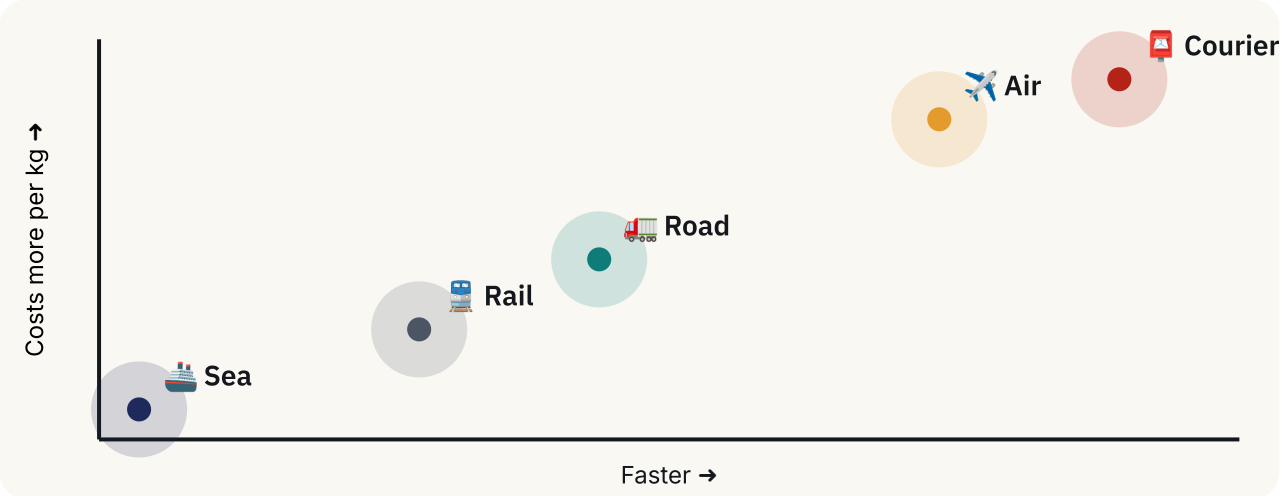
- **Cargo** = goods travelling. **Export** = going out. **Import** = coming in.
- The **shipper** sends; the **consignee** receives.
- A **freight forwarder** organises the trip; a **carrier** (shipping line or airline) actually moves it.
- **Customs** checks goods at the border; a **CHA** (customs broker) does the customs forms for you.
- Shipping is a relay race: every helper passes on the goods *and* the information.

CHAPTER 2

How cargo travels

Ships, planes, trucks, trains and couriers. Each one is good at something different, like choosing between a bus, a taxi and a bicycle.

In this chapter you will learn: the five ways to move cargo, FCL and LCL, what multimodal means, and how to choose.



The trade-off: the faster the trip, the more you usually pay for each kilogram.

🚢 Sea freight

Most of the world's goods travel by sea. Huge ships carry thousands of steel **containers**. Sea is slow but cheap per kilo, so it is perfect for heavy or bulky goods like tea chests, clothes, tyres and furniture.

📦 FCL – Full Container Load

You rent a **whole container** just for your goods. It is packed at your factory, sealed, and opened only by your buyer.

Like hiring a whole van for your house move.

🧩 LCL – Less than Container Load

Your goods **share** a container with other people's goods. A forwarder collects everyone's boxes at a CFS and packs them together (this is called **consolidation**).

Like sending one suitcase on a shared bus.

Rule of thumb: if your goods fill **roughly half a 20-foot container or more**, FCL is often cheaper and safer than LCL. LCL is priced by volume (cubic metres, **CBM**) or weight, whichever is more, called **W/M** (weight or measure; usually 1 CBM is compared with 1,000 kg). Always compare real quotes.

✈️ Air freight

Planes are fast: a few days door to door instead of weeks. But they cost much more per kilo and have tight size limits. Good for urgent, light or valuable things: fashion samples, medicines, spare parts, fresh seafood.

💡 Why a big light box costs more

Airlines charge for the **space** your cargo takes, not just its weight. A box of feathers is light but big! So they work out a **volumetric weight**: length × width × height in centimetres ÷ 6,000. You pay for whichever is bigger, the real weight or the volumetric weight. This is the **chargeable weight**.

Example: a carton 100 × 60 × 30 cm weighing 20 kg. Volume weight = 180,000 ÷ 6,000 = **30 kg**. You pay for 30 kg. (Couriers often divide by 5,000, so check your rate.)

🚛 Road freight

Trucks carry containers and loose cargo between factories, ports, airports and warehouses. In Sri Lanka, almost every shipment uses a truck for the first and last part of the trip (for example, Katunayake to Colombo port). Across continents, trucks also cross borders, such as in Europe.

🚆 Rail freight

Trains move many containers at once over land, cheaper than trucks over long distances and faster than ships on some routes (for example, between China and Europe). Sri Lanka is an island, so our cargo cannot leave by train, but your goods may ride a train at the other end.

📮 Courier (express)

Companies like DHL, FedEx, UPS and Aramex pick up small parcels and deliver them door to door, usually handling customs too. Best for documents, samples and small, urgent parcels.

🔗 Multimodal

Multimodal means using more than one mode on one trip, under one contract: for example *truck → ship → train → truck*. A forwarder organises it so you deal with one company. **Sea-air** is a special mix: ship to a hub like Dubai, then fly onwards, cheaper than all-air and faster than all-sea.

Mode	Good at	Not so good at	Typical Colombo example
🚢 Sea FCL	Lowest cost per kg for big loads; sealed container	Slow (weeks); fixed sailing days; port delays	40' container of garments Colombo → Felixstowe, UK, about 3–5 weeks
🚢 Sea LCL	Small loads without paying for a whole box	Extra handling at CFS; more risk of damage; slower	2 CBM of tea samples to Dubai
✈️ Air	Fast (days); safe; reliable schedules	Expensive; size and dangerous-goods limits	Fashion samples CMB → London Heathrow (LHR)
🚛 Road	Door to door; flexible	Only on land; traffic; weight limits	Container from Biyagama factory to Colombo port
🚆 Rail	Many containers over long land distances	Only where rails go; transfers needed	Rotterdam port to a warehouse in Germany
📮 Courier	Very easy; door to door; customs done for you	Most expensive per kg; small parcels only	A 3 kg box of brochures to a buyer in Paris
🔗 Multimodal	One contract, best mix of speed and cost	More hand-overs to manage	Sea Colombo → Dubai, then air to Europe

⭐ **Top tip**

Transit times in this booklet are typical ranges only. Real times change with the shipping line, the route, transhipment and the season. Always check the carrier's schedule.

Remember!

- Sea = cheap and slow; air = fast and expensive; road = door to door; courier = easy for small parcels.
- **FCL** = your own whole container. **LCL** = sharing a container with others.
- Air freight charges on **chargeable weight**: the bigger of real weight and (L × W × H cm ÷ 6,000).
- **Multimodal** = more than one mode, one contract.

CHAPTER 3

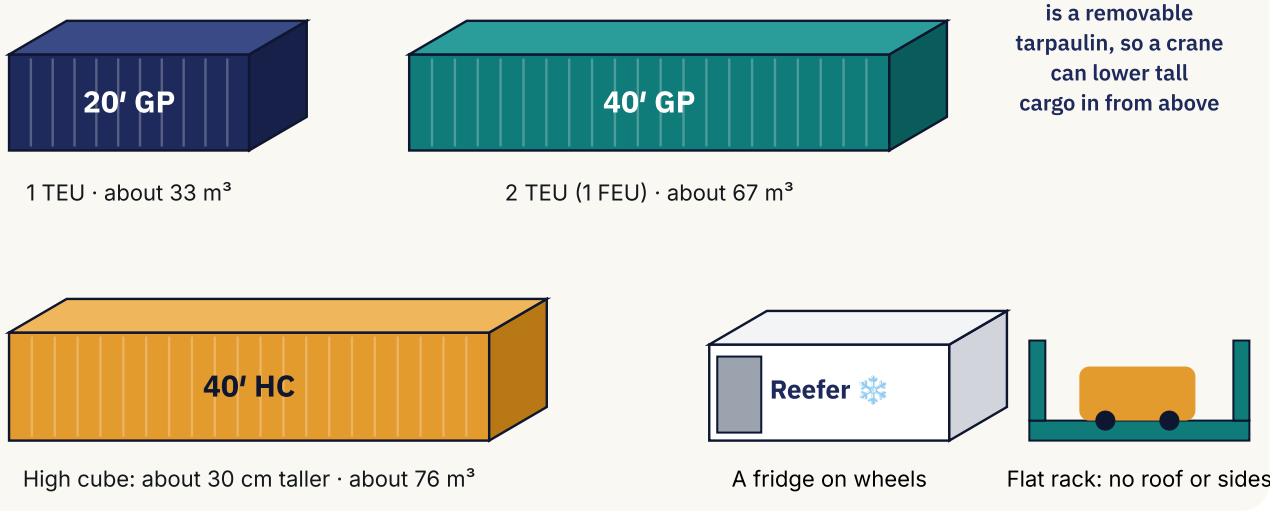
Containers and ULDs

A shipping container is a giant steel lunchbox. It keeps the goods together, safe and dry, and fits every ship, crane and truck in the world.

In this chapter you will learn: the main container types and their sizes, what a TEU is, how to read a container number and a seal, and the boxes and pallets used on planes.

💡 Why containers changed the world

Before containers, workers loaded sacks and boxes onto ships one by one. It took days. Then someone had the clever idea: put everything in **standard-sized boxes**. Now a crane lifts a whole box on or off a ship in a minute or two, and the same box can ride a truck, a ship and a train without ever being opened. It is like a lunchbox that fits every school bag in the world.



The main container family. GP = general purpose (a normal dry box). HC = high cube (taller).

Sizes and how much they hold

All standard containers are 8 feet (2.438 m) wide on the outside. A 20′ is about 6.06 m long and a 40′ is about 12.19 m long. Normal boxes are 8′6″ (2.59 m) tall; high cubes are 9′6″ (2.90 m) tall. The inside figures below are **typical** (from Hapag-Lloyd's published fleet examples). Every box is a bit different, so always read the numbers painted on the door.

Type	Inside length × width × height	Volume	Max payload*	Good for
20GP (20′ dry)	5.90 × 2.35 × 2.39 m	about 33 m³	about 28,000 kg	Heavy goods: tea, tiles, rubber

Type	Inside length × width × height	Volume	Max payload*	Good for
40GP (40′ dry)	12.03 × 2.35 × 2.39 m	about 67 m³	about 26,500–28,800 kg	Bulky goods: garments, furniture
40HC (40′ high cube)	12.03 × 2.35 × 2.70 m	about 76 m³	about 28,600 kg	Light, bulky goods: clothes on hangers, mattresses
20′ reefer	about 5.45 × 2.28 × 2.16 m	about 28 m³	about 27,000–29,000 kg	Chilled or frozen food, fruit, seafood, medicines
40′ HC reefer	about 11.6 × 2.29 × 2.43 m	about 67 m³	about 29,000 kg	Large chilled loads
20′ open top	about 5.90 × 2.35 × 2.34 m	about 32 m³	about 30,000 kg	Tall cargo lowered in by crane
40′ flat rack	about 12.2 × 2.2 m floor, no roof	–	about 40,000 kg or more	Machinery, boats, vehicles, oversized cargo

*Payload = the most cargo you may put in. Gross weight (cargo + box) has its own limit, painted on the door as MAX GROSS. Road weight limits in each country may be lower than the container's limit, so check before you pack it full of something heavy.

Three weights to know

- **Tare**: the empty box (a 20′ is about 2,300 kg).
- **Payload**: the cargo you can add.
- **Gross**: tare + cargo. Never go over the MAX GROSS printed on the door.

Reefers, open tops and flat racks

A **reefer** (refrigerated container) has its own cooling machine. It plugs into power at the port and on the ship. You set the temperature (well below freezing up to about +30 °C on many units), and the box records it. An **open top** has a tarpaulin roof, for tall things like machines that must go in from above. A **flat rack** is just a strong floor with two end walls, for things too big for any box. Open tops and flat racks often count as **out of gauge** (OOG) cargo and cost extra.

TEU: counting containers

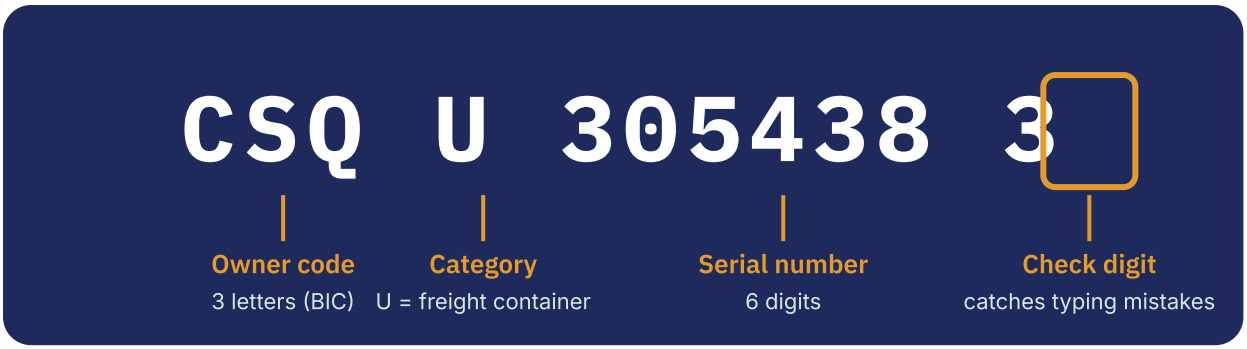
Ports and shipping lines count in **TEU** (twenty-foot equivalent unit). A 20′ box is 1 TEU. A 40′ box is 2 TEU (also called 1 **FEU**, forty-foot equivalent unit). When someone says "this ship carries 20,000 TEU", it could carry 20,000 twenty-footers, or 10,000 forty-footers.



One 20′ plus one 40′ makes 3 TEU.

Reading a container number

Every container has a unique 11-character number painted on its doors, sides and roof. It follows a world standard called **ISO 6346**, and owner codes are registered with the BIC (Bureau International des Containers).



CSQU 305438 3 is the standard's own textbook example. The last digit is worked out from the first ten using a formula.

> 💡 **Think of it like this**
>
> The **check digit** is like a secret handshake. A computer works it out from the other ten characters. If someone types one letter wrong, the handshake fails and the computer says, "That number doesn't exist." Ternary Cargo can use this to catch typing mistakes.

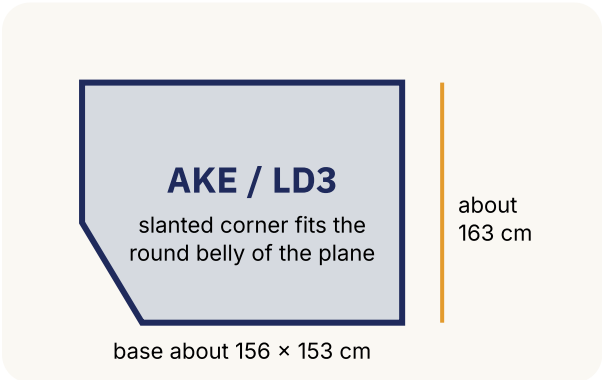
Seals: the container's lock and promise

After an FCL container is packed, the doors are closed with a **seal**: a numbered plastic or steel bolt that can only be removed by cutting it. High-security bolt seals follow the standard **ISO 17712**. The seal number is written on the bill of lading. When the container arrives, the receiver checks that the seal is unbroken *and* the number matches. If it matches, nobody has opened the box on the way.

> ⚠️ **Watch out!**
>
> If a seal is broken, missing or has a different number on arrival, **do not open the container until you have photographed it and told the carrier and your insurer**. Customs may also have opened it for inspection; they put on their own new seal and a record.

Air cargo: ULDs and pallets

Planes do not carry sea containers. They use lightweight aluminium boxes and pallets called **ULDs** (unit load devices), shaped to fit inside the curved body of the aircraft.



The AKE, also called LD3, the most common air container.

ULD	About its size	Max gross
AKE / LD3 container	Base 156 × 153 cm, height 163 cm, about 4.3 m³	1,588 kg
PMC pallet	244 × 318 cm (96 × 125 in)	depends on the aircraft
PAG pallet	224 × 318 cm (88 × 125 in)	depends on the aircraft

Pallets are flat sheets: cargo is built up on them, wrapped in plastic and held down with a net. The cargo must fit a **contour** (shape) that matches the plane's lower deck or main deck. Each airline publishes exact limits.

Remember!

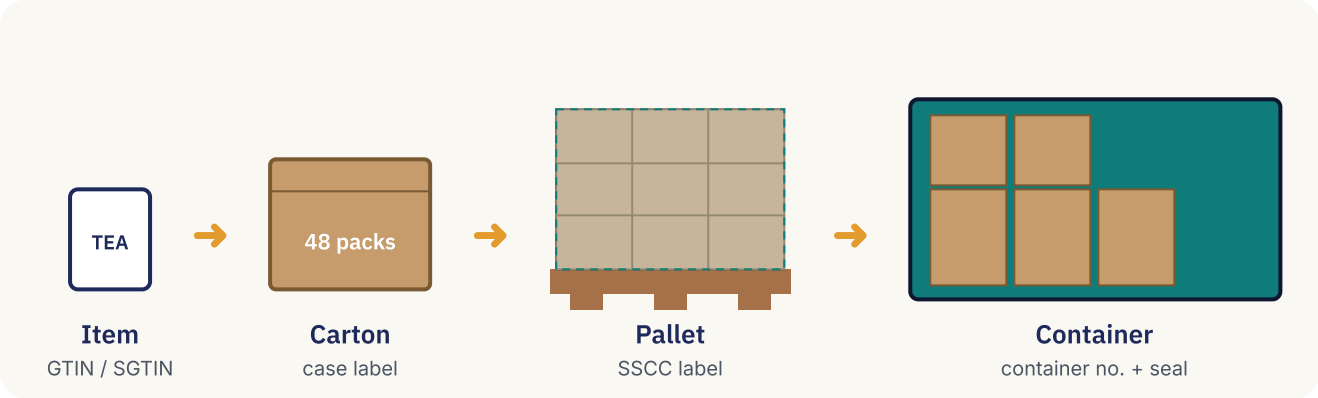
- Containers are standard steel lunchboxes: **20GP**, **40GP**, **40HC** (taller), **reefer** (cold), **open top**, **flat rack**.
- A 20′ holds about 33 m^3; a 40′ about 67 m^3; a 40HC about 76 m^3. Always check the door for the real numbers.
- **1 TEU** = one 20-foot container. A 40′ = 2 TEU.
- Container numbers have 4 letters + 6 digits + 1 check digit, e.g. CSQU 305438 3.
- The **seal number** on the bill of lading must match the seal on the doors when the box arrives.
- Planes use **ULDs**: the **AKE/LD3** container (max 1,588 kg) and PMC/PAG pallets.

CHAPTER 4

Packing, labels and barcodes

Good packing keeps goods safe. Good labels let everyone find and count them. Barcodes let a computer do the counting in a blink.

In this chapter you will learn: cartons and pallets, how to palletise, shipping marks, the GS1 numbers GTIN, SGTIN and SSCC, barcodes and scanning, and what packing lists and ASNs are.



Like Russian dolls: items go into cartons, cartons onto pallets, pallets into a container. Each level has its own label.

Cartons

Most goods travel in **corrugated cardboard cartons** (the wavy layer inside makes them strong but light). Heavy goods need **double-wall** or triple-wall cartons. A good carton is full (no empty space for things to rattle), taped well, and not too heavy for one person to lift safely.

Shipping marks and handling symbols

Each carton is printed with **shipping marks**, so anyone can match it to the paperwork: the buyer's name or code, destination, order number and carton number such as `HAM / PO-5521 / 14 of 240`. Pictures called **handling symbols** tell workers how to treat it (they follow the standard ISO 780):

⬆️⬆️
This way up
Keep the arrows pointing up.

🥂
Fragile
Handle with care.

☂️
Keep dry
Protect from rain and damp.

Pallets and palletising

A **pallet** is a flat wooden or plastic platform. A forklift slides its forks underneath and moves a whole stack of cartons in one go. Putting cartons on pallets is called **palletising**. Common sizes are **1200 × 1000 mm** (common in Asia and the UK), **1200 × 800 mm** (the "Euro pallet") and **1100 × 1100 mm**.

1. 🧱 **Heavy at the bottom**
   Put the heaviest cartons on the bottom layer, lighter ones on top.

2 **🧩 Stack neatly**

Use a column stack (cartons exactly on top of each other, strongest) or an interlocking pattern (like bricks, more stable). Do not let cartons hang over the edge.

3 **🎁 Wrap and strap**

Wrap the stack in stretch film, add corner boards, and strap it to the pallet.

4 **🏷️ Label it**

Stick two identical pallet labels with the **SSCC** barcode: one on a short side and one on a long side, roughly 400–800 mm above the floor.

5 **⚖️ Weigh and measure**

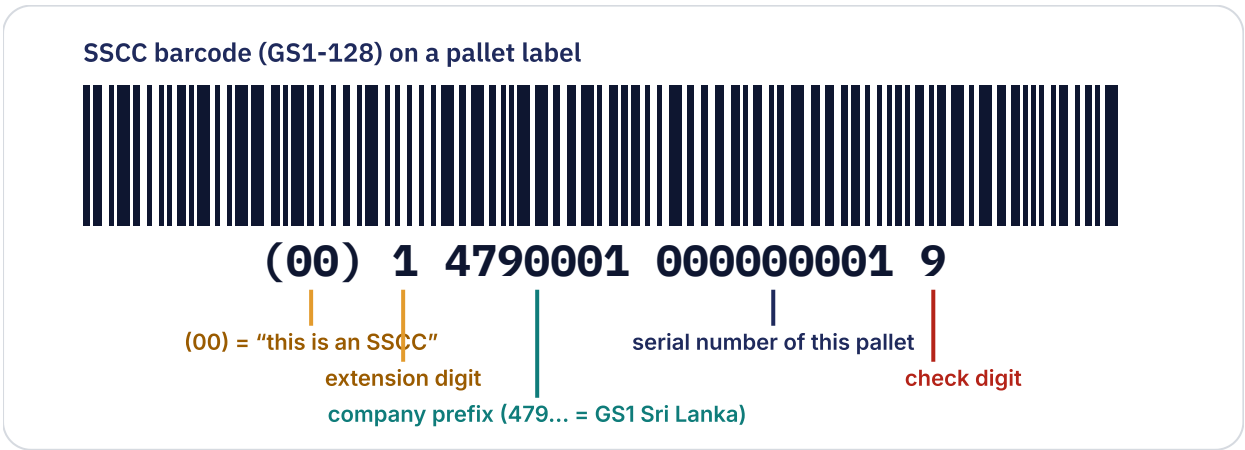
Record the pallet's height and gross weight. These go on the packing list and into the load plan.

> **⚠️ Wooden pallets need a passport too**
>
> Wooden pallets and crates for export must usually be **heat-treated or fumigated** and stamped with the **ISPM 15** mark (it shows "HT" for heat-treated). Many countries reject or destroy untreated wood because it can carry insects.

GS1 numbers: names for things

GS1 is a worldwide organisation that hands out barcode numbers so that no two things in the world get the same one. Sri Lankan companies get numbers starting with **479** from GS1 Sri Lanka. There are three numbers you will meet:

Number	What it names	Think of it like…	Example
GTIN Global Trade Item Number	A **type** of product, e.g. "Ceylon black tea, 100 bags". Every box of this product has the same GTIN.	The name of a book	4790001234561 (a 13-digit GTIN, printed as an EAN-13 barcode)
SGTIN Serialised GTIN	One **exact** item or carton: the GTIN plus a unique serial number.	The book's name + a library copy number	(01) 04790001234561 (21) 000123
SSCC Serial Shipping Container Code	One **logistic unit**: a pallet, a parcel or a mixed carton that travels as one.	A luggage tag for one suitcase	(00) 1 4790001 000000001 9 (18 digits)



The SSCC is the only must-have item on a GS1 pallet label. If a pallet is broken up and rebuilt, it gets a new SSCC.

💡 What are the numbers in brackets?

The little numbers in brackets, like **(00)**, **(01)** and **(21)**, are called **Application Identifiers**. They tell the scanner what kind of number comes next, like the words “Name:” and “Age:” on a form. (00) means SSCC, (01) means GTIN, (21) means serial number.

Barcodes and scanning

A barcode is a number written in stripes or squares that a scanner can read in a split second, with no typing mistakes.

Barcode	Looks like	Used for
EAN-13 / UPC	Stripes with 13 (or 12) digits	Products in shops (GTIN)
GS1-128	Long stripes with (brackets)	Cartons and pallets: SSCC, batch, dates
QR code / GS1 DataMatrix	A square of little dots	Lots of data in a small space; phone scanning; serials

When workers **scan** each pallet as it is loaded and again when it is unloaded, the computer can tell instantly if something is missing, extra or in the wrong container. Ternary Cargo’s phone scanner works even with no internet and catches up later.

Packing list and ASN

The **packing list** is the list of everything in the shipment: every pallet and carton, what is inside, the marks, the weights and the sizes. It does *not* show prices (that is the invoice’s job). Customs, the forwarder, the warehouse and the buyer all use it.

An **ASN** (Advance Shipping Notice, also called a despatch advice) is the same information sent **electronically to the buyer before the goods arrive**. When the buyer scans each SSCC on arrival, the system ticks it off against the ASN. Anything not on the list, or missing, is flagged straight away.

🇱🇰 A Sri Lankan example

Lanka Knit Apparel packs 2,400 T-shirts into 100 cartons (24 shirts each) and stacks them on 10 pallets. Each carton gets an SGTIN label and each pallet an SSCC label. The packing list shows: pallet 1, SSCC ...0019, cartons 1–10, 240 shirts, 186 kg, 120 × 100 × 150 cm, and so on. The UK buyer receives this as an ASN the day the ship sails.

Remember!

- Items go in **cartons**, cartons on **pallets**, pallets in **containers**.
- Mark every carton with shipping marks and handling symbols; heavy at the bottom; no overhang; wrap and strap.
- Export wood packaging needs the **ISPM 15** stamp.
- **GTIN** = which product. **SGTIN** = which exact item. **SSCC** = which pallet or parcel.
- The **packing list** says what is in the shipment (no prices). The **ASN** sends it to the buyer ahead of time so they can scan and check.

CHAPTER 5

Loading the container

Packing a container is like packing a car boot for a long trip: heavy things low and in the middle, nothing loose, and never more than it can carry.

In this chapter you will learn: how a load plan works, volume and weight utilisation, centre of gravity, stacking rules, dangerous goods, securing cargo, and the VGM weight rule.

The load plan

A **load plan** is a drawing that shows where every pallet or carton goes inside the container, *before* the truck arrives. It answers three questions:

1. **Does it fit?** (space, in cubic metres)
2. **Is it too heavy?** (weight, in kilograms)
3. **Is it safe?** (balance, stacking, dangerous goods)

> 💡 **Think of it like this**
>
> A load plan is like playing a game of Tetris in 3D before you start, so you know which pieces go in first. Remember: the first thing you put in is the last thing out! If a container stops at two buyers, load the second buyer's goods first (at the front) and the first buyer's goods last (by the door).

Utilisation: how full is the box?

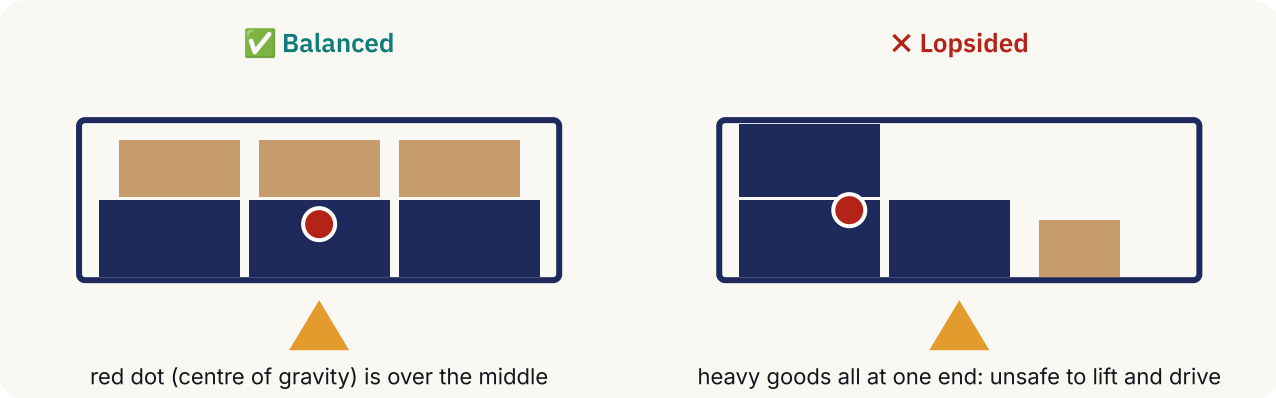
Volume utilisation = space used ÷ space inside. **Weight utilisation** = cargo weight ÷ maximum payload. You can "cube out" (run out of space first, common with clothes) or "weigh out" (hit the weight limit first, common with tea, tiles or granite).

> 🇱🇰 **A Sri Lankan example**
>
> Ceylon Leaf Teas fills a 20GP with 20 pallets of tea. Space used is about 25 m^3 of 33 m^3 (76%). Weight is 18,000 kg of a 28,000 kg payload (64%). Room to spare on both. Lanka Knit's T-shirts fill a 40HC to 92% of its space but only 35% of its weight: they cube out.

Why you never fill 100%

Pallets and cartons are not exactly the right shape for the box, forklifts need room to work, and you need a little space for securing materials. Real loads of pallets often fill around 70–90% of the space. A 3D load planner shows you the real number.

Weight and balance



Keep the centre of gravity (the balance point) near the middle of the container, both along its length and across its width.

- **Spread weight evenly** along the floor. Never put all the heavy cargo at one end or one side.
- **Heavy at the bottom**, light on top.
- Remember the truck: on the road, too much weight over one axle is dangerous and may be illegal. Road weight limits differ in every country.
- Never go over the container's **MAX GROSS** weight.

Stacking

Some cargo can carry weight on top; some cannot. Labels like **DO NOT STACK** or a top-load limit (for example "max 500 kg on top") must be obeyed. Crushed bottom cartons are one of the most common causes of damage claims.

Securing the load

A ship rolls and a truck brakes hard. Cargo that can slide will smash. Fill gaps and tie things down using **dunnage** (wood or packing to fill space), **air bags** (inflatable cushions pushed into gaps), **lashing** straps to the container's lashing rings, and **blocking and bracing** with timber. The international guide for this is the **CTU Code** (IMO/ILO/UNECE Code of Practice for Packing of Cargo Transport Units).

Dangerous goods (hazmat)

Some cargo can catch fire, explode, poison or burn: paints, perfumes, gas cylinders, many chemicals, and **lithium batteries** (in phones, laptops and power banks). These are **dangerous goods** (DG). They are allowed, but with strict rules:

Rule book	For	What it means
IMDG Code	Sea	How to pack, label, document and separate dangerous goods on ships
IATA DGR	Air	Airline rules; some goods are banned on planes or only on cargo planes

Every dangerous good has a **UN number** (for example UN3480 for lithium-ion batteries on their own) and belongs to one of **9 classes** (1 explosives, 2 gases, 3 flammable liquids, 4 flammable solids, 5 oxidisers, 6 toxic and infectious, 7 radioactive, 8 corrosive, 9 miscellaneous). The shipper must sign a **dangerous goods declaration**, use approved packaging, put on the diamond labels, and keep some classes apart from each other (**segregation**).

⚠️ **Watch out!**

Never hide or mis-describe dangerous goods. Undeclared DG have caused fires on container ships. It is illegal and you can be fined or prosecuted.

VGM: weigh it before it sails

Since **1 July 2016**, a world safety rule (SOLAS, Chapter VI, Regulation 2) says every packed export container must have a **Verified Gross Mass (VGM)** before it is loaded on a ship. No VGM, no loading. The shipper can get it in two ways:

⚖️ **Method 1**

Weigh the whole packed and sealed container on a certified weighbridge.

🧱 **Method 2**

Weigh all the cargo, pallets, packing and securing material, then add the container's tare weight (painted on the door), using an approved method.

The VGM is sent to the shipping line before the **VGM cut-off** (a deadline, usually a day or more before the ship arrives).

⭐ **Cut-offs are like the school bell**

Each port and line has several **cut-offs**: the **CY cut-off** (last time to deliver the container to the terminal), the **SI cut-off** (last time to send shipping instructions for the bill of lading), the **VGM cut-off**, and the **customs cut-off**. Miss one and your container may be **rolled** (left behind for the next ship).

Air loading

At the airport, the airline or its ground handler "builds up" ULDs: they stack cartons on pallets or into AKE containers to fit the aircraft's shape and weight limits, then a load planner balances the whole plane.

Remember!

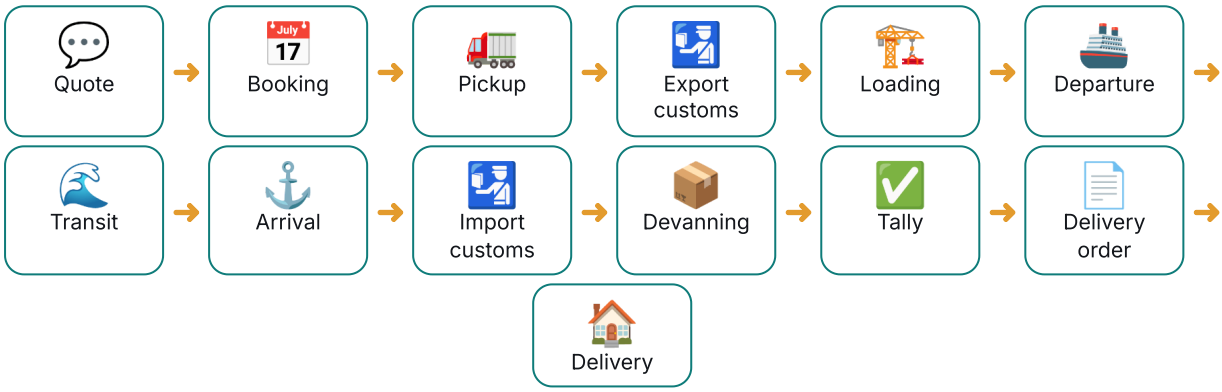
- Plan before you pack: does it **fit**, is it too **heavy**, is it **safe**?
- Keep the **centre of gravity** near the middle. Heavy at the bottom. Spread the weight.
- Obey **DO NOT STACK**. Fill gaps and tie everything down.
- Dangerous goods need a **UN number**, labels and a signed **DG declaration** (IMDG for sea, IATA DGR for air).
- No **VGM**, no loading. Meet every **cut-off** or risk being rolled.

CHAPTER 6

The shipment journey, step by step

Let's follow one container of Ceylon tea from a warehouse in Colombo to a shop's warehouse in Hamburg, Germany. Then we will see what is different by air.

In this chapter you will learn: the 14 steps of a typical shipment, who does each one, and which document or event belongs to it.



1 💬 Quote

Ceylon Leaf Teas asks its forwarder: "How much to send 20 pallets of tea, Colombo (LKCMB) to Hamburg (DEHAM), FCL 20GP?" The forwarder sends a **quotation** with the freight rate, surcharges, transit time and validity date. *Who:* exporter and forwarder.

2 📅 Booking

The exporter accepts. The forwarder **books** space on a ship with a shipping line. The line sends a **booking confirmation** with the vessel, voyage, cut-off times and a release order to collect an empty container. *Who:* forwarder and carrier.

3 🚛 Pickup and packing (stuffing)

A haulier collects an empty 20GP from the container depot, drives it to the exporter, and the exporter **stuffs** (packs) it, following the load plan. The doors are closed with a **seal**. Everyone notes the container and seal numbers. For LCL, the exporter instead delivers cartons to the forwarder's CFS.

4 🛂 Export customs

The exporter's CHA lodges an export **CUSDEC** (customs declaration) in **ASYCUDA World** with the invoice and packing list. Sri Lanka Customs may check papers or inspect the goods, then give export clearance. The exporter also sends the **VGM** and **shipping instructions** (SI) to the line.

5

🏗️ Gate-in and loading

The truck takes the container to the port terminal before the **CY cut-off**. At the gate it is checked and recorded (an **EIR**, equipment interchange receipt). A gantry crane lifts it onto the ship.

6

🚢 Departure

The ship sails: this is the **ATD** (actual time of departure). The line issues the **bill of lading**. The forwarder sends the buyer the documents and an ASN.

7

🌊 Transit and transhipment

The container is at sea. Often it changes ships at a hub port; this is **transhipment**. Colombo itself is a big transhipment hub where containers from India and Bangladesh swap ships. Tracking updates show each milestone and the **ETA** (estimated time of arrival).

8

⚓ Arrival

Before arrival, the line files the cargo **manifest** with destination customs, and the line or forwarder sends the buyer an **arrival notice**. The ship berths, the container is discharged and waits in the terminal yard. The **free time** clock starts (see Chapter 10).

9

🛂 Import customs

The buyer's customs broker lodges the import declaration, pays duty and VAT, and gets the goods released. (When Sri Lanka imports, this is the import CUSDEC in ASYCUDA World.)

10

📄 Delivery order and release

The buyer pays the freight and local charges and hands over the original bill of lading (or the line accepts a telex release). The line issues a **delivery order** (DO): permission to collect the container.

11

📦 Unloading (devanning)

A truck takes the container to the buyer's warehouse or a CFS. The seal is checked and cut, and the cargo is taken out. Unloading a container is called **devanning** or **destuffing**.

12

✅ Tally and verification

Workers **tally** (count) every package, scan the SSCC labels against the ASN and packing list, weigh where needed, and note any damage with photos. A supervisor signs off: everything matches, or a **discrepancy report** is written.

13

🏠 Delivery and empty return

The goods are put away or delivered to the final address, and the receiver signs a **proof of delivery** (POD). The empty container goes back to the line's depot before the detention free time ends.

14

💰 Invoicing and closing

The forwarder sends its invoices, everyone is paid, and the job is closed. Now it's time for the next shipment!

What changes by air?

Step	Sea	Air
Booking	Space on a vessel, container released	Space on a flight; airline gives an air waybill number
Packing	Packed into a container at the factory or CFS	Delivered loose to the airline's cargo terminal; built up on ULDs there
Security	Container sealed	Cargo is security screened (or comes from a secure, known shipper)
Document	Bill of lading	Air waybill (AWB)
Time	Weeks	Usually 1–5 days
Arrival	Container to yard; free days	Cargo to the airline's import shed; storage often charged after a short free period

🇱🇰 A Sri Lankan example

Lanka Knit sends 300 kg of urgent samples by air from Bandaranaike International Airport (**CMB**) to London Heathrow (**LHR**). The forwarder books a flight, the cartons go to the airport cargo terminal, the CHA lodges the export CUSDEC, the cargo is screened and loaded, and the buyer's broker clears it in London two days later.

Remember!

- The big steps: **quote → booking → pickup → export customs → loading → departure → transit → arrival → import customs → delivery order → devanning → tally → delivery**.
- Goods and papers travel separately. Each step has a document or an event.
- **Transhipment** = changing ships on the way. Colombo is a hub for this.
- **Devanning** = unloading a container. **Tally** = counting and checking what came out.
- You need a **delivery order** before you can collect cargo at the destination.

CHAPTER 7

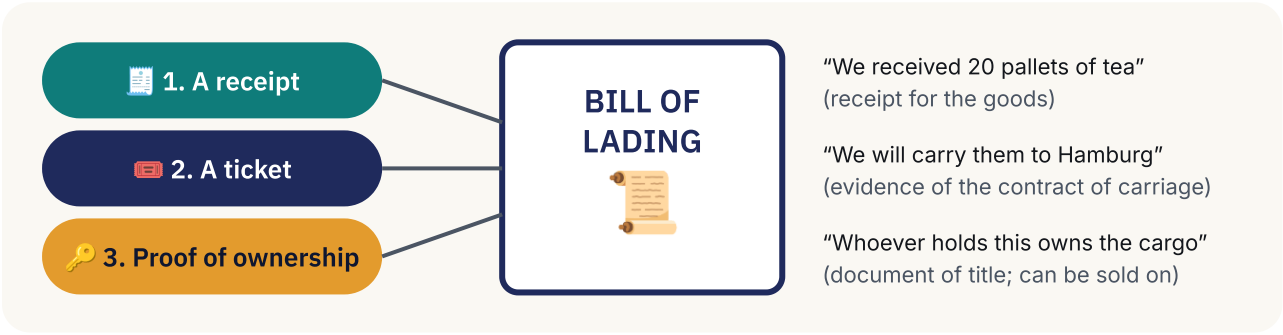
The papers that travel with cargo

Goods never travel alone. A stack of documents goes with them, mostly by email or computer today. Each paper has one job.

In this chapter you will learn: what each key document is for, who makes it, and the difference between master and house documents.

The big one: the bill of lading

The **bill of lading** (BL or B/L) is issued by the shipping line (or the forwarder) when it takes your goods. It does three jobs at once:



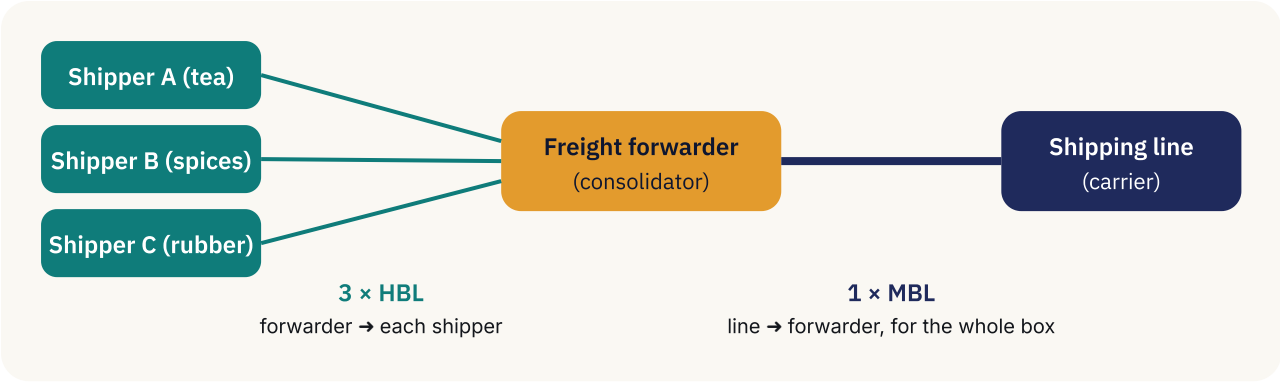
A bill of lading = a receipt + a ticket + proof of ownership.

Because a **negotiable** ("to order") original BL is proof of ownership, whoever holds the original can collect the cargo. That is why banks hold it until the buyer pays. Other kinds:

- **Original BL**: usually issued as a set of three paper originals. Any one can be used to collect the goods.
- **Telex release / surrendered BL**: the originals are handed back at origin, and the line releases the cargo at destination without paper.
- **Sea waybill**: a receipt and contract only, not proof of ownership. The named consignee just collects the goods. Quick and easy when buyer and seller trust each other.
- **Electronic BL (eBL)**: a digital BL on an approved platform.

MBL and HBL: master and house

When a forwarder joins several customers' goods together, there are two layers of bills:



The line issues one Master BL to the forwarder. The forwarder issues a House BL to each of its customers.

	Master (MBL)	House (HBL)
Issued by	The shipping line	The freight forwarder (or NVOCC)
Issued to	The forwarder	The real shipper (your company)
Covers	The whole container or consolidation	One customer's goods
Used to collect cargo from	The shipping line	The forwarder's agent at destination

The air waybill: MAWB and HAWB

The **air waybill** (AWB) is air freight's version of the BL. It is a receipt and a contract, but **not** proof of ownership, so it cannot be sold on. It has an 11-digit number: the first 3 digits are the airline's code (SriLankan Airlines' prefix is 603). Just like sea, there is a **Master AWB** (airline to forwarder) and a **House AWB** (forwarder to each shipper).

Every document in one place

Commercial invoice

Seller → buyer · needed by customs at both ends

The bill for the goods: who sells, who buys, what, how many, price per unit, total, currency, Incoterm and HS codes. Customs use it to work out duty.

Proforma invoice

Seller → buyer, before the sale

A "preview" invoice used to agree the deal, open a letter of credit or get an import licence.

Packing list

Seller

What is in each pallet and carton, with weights, sizes and marks. No prices.

Shipping instructions (SI)

Shipper → carrier

The details the line needs to print the BL: shipper, consignee, notify party, cargo description, container and seal numbers.

VGM declaration

Shipper → carrier

The verified gross mass of each container, signed.

CUSDEC (customs declaration)

CHA → Sri Lanka Customs, in ASYCUDA World

Sri Lanka's customs declaration form, for both exports and imports. It lists the goods, HS codes, values and the taxes due.

Bill of lading / sea waybill

Carrier → shipper

See above. Sea and multimodal.

Air waybill (AWB)

Airline or forwarder → shipper

See above. Air.

Certificate of origin (CO)

Issued by an authority or chamber of commerce

Proves where the goods were made. A **preferential** CO can lower the buyer's import duty under a trade deal. In Sri Lanka, check with the Department of Commerce or your chamber of commerce which form your buyer's country needs.

Insurance certificate

Insurer → whoever must insure

Proof the cargo is insured, for how much and against which risks. Required under CIF and CIP.

Phytosanitary / health / fumigation certificates

Government agencies

Proof that plants, food or wood are free of pests or disease, when the buyer's country asks for it.

Dangerous goods declaration

Shipper

Required for any hazardous cargo (Chapter 5).

Cargo manifest

Carrier → customs

The list of everything on a ship or plane. In Sri Lanka, sea manifests must be lodged electronically through **ASYHUB** (since 7 May 2025).

Arrival notice

Carrier or forwarder → consignee

"Your cargo arrives on this date. Here are the charges and the free time."

Delivery order (DO)

Carrier or forwarder → consignee

Permission for the consignee (or their truck) to collect the cargo from the terminal or CFS. Issued after charges are paid and the BL is surrendered.

EIR (equipment interchange receipt)

Terminal or depot

Records the container's condition each time it changes hands at a gate.

Tally sheet and proof of delivery (POD)

Warehouse; receiver

The count of what arrived, and the receiver's signature saying "I got it".

⭐ Top tip

The same facts appear on many papers: weights, number of packages, marks, the description. **They must match exactly.** If the invoice says 240 cartons and the BL says 204, banks may refuse payment and customs may hold the goods. Ternary Cargo makes every document from one set of data so they always agree.

Remember!

- The **bill of lading** = receipt + ticket + proof of ownership. Whoever holds the original (to-order) BL can collect the goods.
- **MBL**: line → forwarder. **HBL**: forwarder → each shipper. Same idea for air: **MAWB** and **HAWB**.
- An **air waybill** and a **sea waybill** are not proof of ownership.
- The **commercial invoice** shows prices; the **packing list** shows contents; the **CUSDEC** declares to Sri Lanka Customs.
- The **arrival notice** says it is coming; the **delivery order** lets you collect it.
- All documents must tell the same story.

CHAPTER 8

Incoterms® 2020: who pays for what

When you buy a bike online, you check: does the price include delivery? Incoterms are the same question for world trade, answered with a three-letter code.

In this chapter you will learn: all 11 Incoterms 2020 rules, where the costs and the risk pass from seller to buyer, and which ones to use for containers.

Incoterms® are rules written by the **International Chamber of Commerce (ICC)**. The current version is **Incoterms 2020**. Each rule is a three-letter code plus a named place, like **FOB Colombo** or **DAP Hamburg**. It tells the seller and buyer three things:

1. **Costs**: who pays for each part of the journey.
2. **Risk**: the exact point where, if the goods are lost or damaged, it becomes the buyer's problem instead of the seller's.
3. **Tasks**: who does export and import customs, who books transport, who insures.

> 💡 **Think of it like this**
>
> Imagine passing a glass of water to a friend. Until your friend's hand is holding it, if it drops, it's your fault. Once they have it, it's theirs. The Incoterm says exactly **where the hand-over happens**.

> ⚠️ **Two common mix-ups**
>
> Incoterms do **not** say when ownership passes or how and when the buyer pays. Those go in the sales contract. And with the four **C** rules, the seller pays for the main trip but the **risk passes early**, at the start of the trip. Paying and risk are not the same thing!

How to read the pictures

■ **Seller pays** ■ **Buyer pays** □ Not required by the rule (whoever carries the risk usually insures) ▲ where the **risk** passes to the buyer.

Simplified: for the C rules we assume the named place is the destination port; for the D rules, the buyer's premises. Destination port charges under C rules are the buyer's unless the seller's freight contract includes them.

The 7 rules for any kind of transport

EXW Ex Works

Any mode

The seller just gets the goods ready at its own premises. The buyer does everything else, even loading the truck and export customs. Lowest effort for the seller; hard for a foreign buyer, because they must clear Sri Lankan export customs. Often FCA is a better choice.



FCA Free Carrier

Any mode

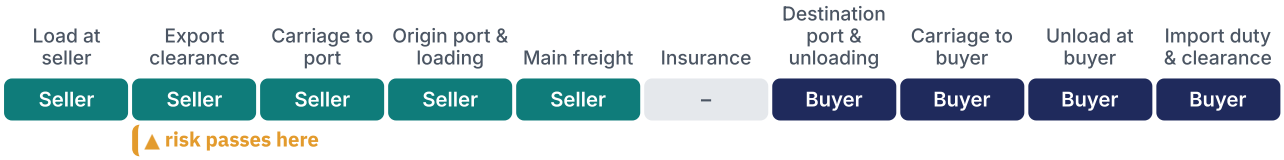
The seller clears export and hands the goods to the buyer's carrier at a named place (for example its factory, loaded onto the truck, or the forwarder's warehouse, ready for unloading). Shown here: at the seller's premises. **The best choice for most container and air shipments where the buyer arranges freight.**



CPT Carriage Paid To

Any mode

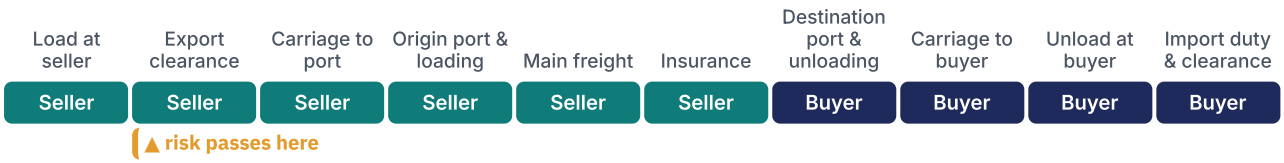
The seller pays the freight to the named destination, but the risk passes as soon as the goods are handed to the **first carrier** (here: the truck at the factory).



CIP Carriage and Insurance Paid To

Any mode

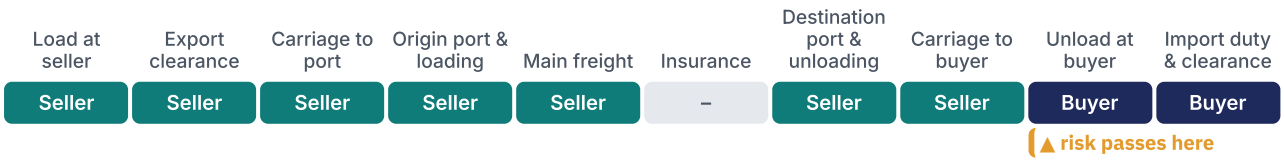
Like CPT, plus the seller must buy insurance for the buyer. Under Incoterms 2020 this must be high cover, **Institute Cargo Clauses (A)** ("all risks"), at least 110% of the contract value, unless agreed otherwise.



DAP Delivered at Place

Any mode

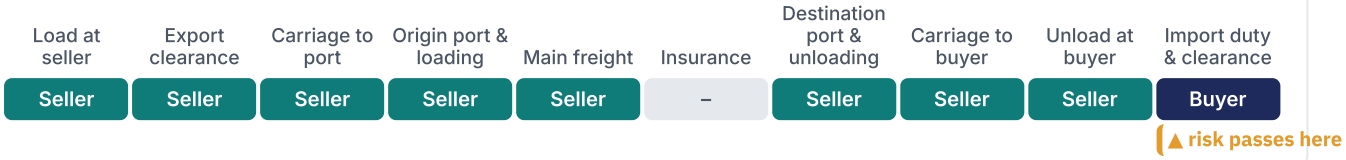
The seller delivers to the named place (for example the buyer's warehouse), ready for unloading. The buyer unloads and does import customs and pays duty.



DPU Delivered at Place Unloaded

Any mode

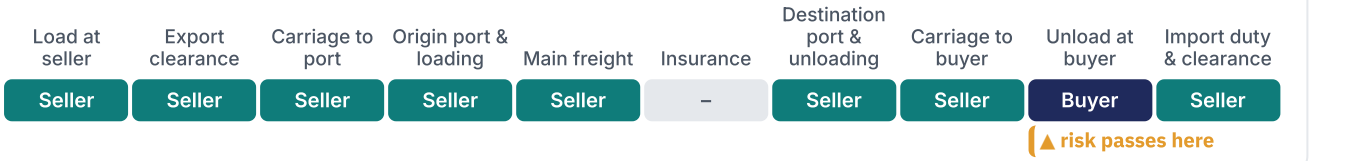
Like DAP, but the seller also **unloads** the goods. The only rule where the seller must unload. (It replaced the old DAT rule in 2020.)



DDP Delivered Duty Paid

Any mode

The seller does everything, including import customs, duty and taxes in the buyer's country. Most work and risk for the seller. Tricky if the seller cannot register for import VAT there.



The 4 rules for sea and inland waterway only

FAS Free Alongside Ship

Sea only

The seller clears export and places the goods **next to** the ship at the named port. Used for bulk cargo and very big items, not containers.



FOB Free On Board

Sea only

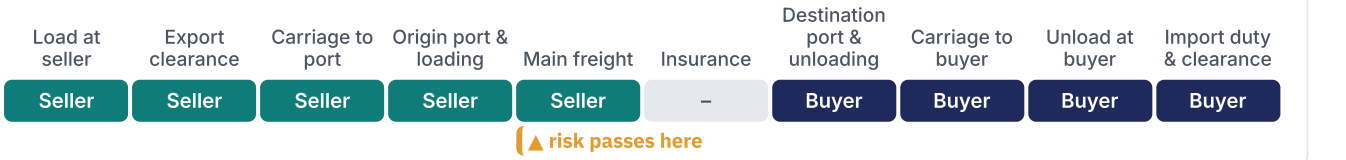
The seller clears export and loads the goods **on board** the buyer's chosen ship at the named port, e.g. **FOB Colombo**. Very common, but for containers (which are handed over at the terminal, not on board) the ICC suggests **FCA** instead.



CFR Cost and Freight

Sea only

The seller pays the sea freight to the destination port, but risk passes when the goods are on board at the origin port.



CIF **Cost, Insurance and Freight** Sea only

Like CFR, plus the seller insures. Under CIF the minimum cover is **Institute Cargo Clauses (C)** (more limited), unless agreed otherwise. For containers, the ICC suggests **CIP** instead.



The family groups

Letter	Means	Rules	Easy way to remember
E	Seller makes goods available at its door	EXW	E = "Exit at the factory"
F	Seller hands over to the buyer's carrier; buyer pays the main trip	FCA, FAS, FOB	F = "Free of the main freight"
C	Seller pays the main trip, but risk passes at origin	CPT, CIP, CFR, CIF	C = "Carriage paid, but Careful: risk passed early"
D	Seller delivers to the destination and carries the risk until then	DAP, DPU, DDP	D = "Delivered"

🇱🇰 A Sri Lankan example

Ceylon Leaf Teas sells tea **FCA Colombo (our warehouse)**. The German buyer's forwarder books the ship. Ceylon Leaf Teas clears export, loads the container on the buyer's truck, and the risk passes there. Lanka Knit sells T-shirts **CIF Felixstowe**: it pays freight and insurance to the UK port, but if the ship has a storm in the Arabian Sea, it is the UK buyer who claims on the insurance.

Remember!

- Incoterms 2020 has **11 rules**: 7 for any mode (EXW, FCA, CPT, CIP, DAP, DPU, DDP) and 4 for sea only (FAS, FOB, CFR, CIF).
- Always write the code **and a place**, plus the version: "FCA Colombo, Incoterms® 2020".
- In **C** rules, the seller pays the freight but the **risk passes at origin**.
- For containers, prefer **FCA, CPT, CIP** over FOB, CFR, CIF.
- **CIP** needs ICC (A) insurance; **CIF** needs ICC (C) minimum.
- Incoterms don't cover payment or ownership: put those in the contract.

CHAPTER 9

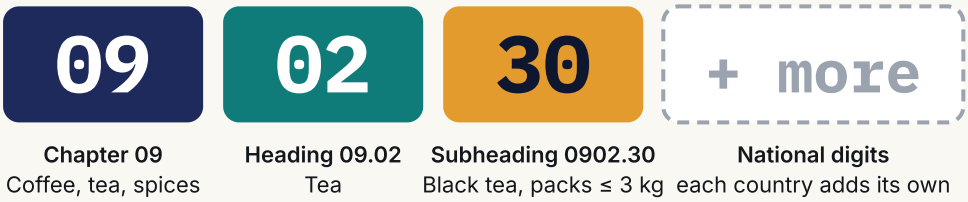
Customs and duty basics

Customs are the gatekeepers of a country. They check what comes in and goes out, keep out dangerous or banned things, and collect taxes.

In this chapter you will learn: what HS codes are, how customs value goods, how duty and VAT are worked out, and how Sri Lanka Customs' systems ASYCUDA World and ASYHUB fit in.

HS codes: a name tag every customs office understands

There are millions of different products. To treat them fairly, almost every country uses the same numbering system: the **Harmonized System** (HS), run by the **World Customs Organization** (WCO). The first 6 digits are the same worldwide. Countries add more digits for their own tariff lines.



Reading an HS code from big group to small: chapter → heading → subheading.

Product	HS (6 digits)	Means
Ceylon black tea in tea bags (packs up to 3 kg)	0902.30	Black tea, in immediate packings ≤ 3 kg
Black tea in bulk	0902.40	Other black tea
Cotton T-shirts, knitted	6109.10	T-shirts, singlets and vests, of cotton, knitted
Desiccated coconut	0801.11	Coconuts, desiccated

⚠️ Watch out!

The wrong HS code can mean paying the wrong duty, fines, or goods being held. The full Sri Lankan codes and rates are in the **Sri Lanka Customs National Imports Tariff Guide**. Always check the latest version, or ask your CHA.

Customs value: what are the goods worth?

Duty is usually a percentage of the goods' value. Most countries use the WTO valuation rules, where the main method is the **transaction value**: the price actually paid, with some costs added. Sri Lanka works out import duty on the **CIF value** (cost of the goods + insurance + freight to Sri Lanka), converted to rupees.

Duty and taxes

Customs duty is a tax on imported goods. Countries also charge **VAT** (value added tax) on imports. In Sri Lanka, Customs collects several levies at import time. The ones you will meet most are:

Levy	What it is (simply)
Customs duty	A % of the CIF value (or a fixed amount per unit) set in the tariff for each HS code
PAL	Port and Airport Development Levy, a % of the CIF value
Cess	Export Development Board cess on certain goods
Excise duty	Extra tax on some goods, such as vehicles, alcohol and tobacco
VAT	18% standard rate since 1 January 2024
SSCL	Social Security Contribution Levy, generally 2.5%

The tariff guide gives the formula for the VAT base: **(CIF value + 10% of CIF + duty + cess + PAL + excise + surcharge) × VAT rate**. SSCL uses the same base.

> 💡 **Let's do the maths (made-up rates)**
>
> A toy worth **LKR 100,000** CIF is imported. Pretend the duty is 10% and PAL is 5%. (These are made-up rates to show the maths, not real ones.)
> Duty = 10,000. PAL = 5,000.
> VAT base = 100,000 + 10,000 (the extra 10%) + 10,000 + 5,000 = 125,000.
> VAT = 125,000 × 18% = **22,500**. SSCL = 125,000 × 2.5% = **3,125**.
> Total taxes = 10,000 + 5,000 + 22,500 + 3,125 = **LKR 40,625**.

Exports: most Sri Lankan exports pay no customs duty, but they still need an export declaration, and some goods carry an export cess or need special permits.

Sri Lanka Customs systems

🖥️

ASYCUDA World

The computer system Sri Lanka Customs uses for customs declarations. Your CHA lodges the **CUSDEC** here, for imports and exports.

Built by UNCTAD

🌐

ASYHUB

The platform for **cargo manifests**. Since 7 May 2025, shipping lines, agents, forwarders and NVOCCs must lodge sea manifests only through ASYHUB.

Manifests: import, export, transhipment

👔

CHA

A **Customs House Agent**: the licensed person or company who prepares and lodges declarations for you.

Like a lawyer for your goods at the border

Inward sea cargo manifests must reach Customs electronically **72 hours before the ship arrives** at the first Sri Lankan port (or when it leaves the last foreign port, if the trip is shorter than 72 hours). After a declaration is lodged, Customs decides how closely to check it: some are released on paperwork only, others need a document check or a physical **examination** of the goods.

The customs steps, simply



⭐ **Top tip**

Other agencies may also need to approve some goods before Customs releases them, for example food, plants, medicines, chemicals or electronics. Ask your CHA early if your product needs a licence or a certificate.

Remember!

- Every product has an **HS code**. The first 6 digits are the same worldwide.
- Sri Lanka calculates import taxes on the **CIF value** in rupees.
- Import levies can include duty, PAL, cess, excise, **VAT 18%** and **SSCL 2.5%**. Check the current tariff guide.
- Declarations (**CUSDEC**) go into **ASYCUDA World**; sea manifests go into **ASYHUB**.
- A **CHA** does this paperwork for you.

CHAPTER 10

Money matters

Every step of the journey costs something. Knowing the names of the charges, and who pays them, saves real money.

In this chapter you will learn: how freight is priced, the common surcharges, landed cost, demurrage and detention, and the main ways to pay: TT, LC and others.

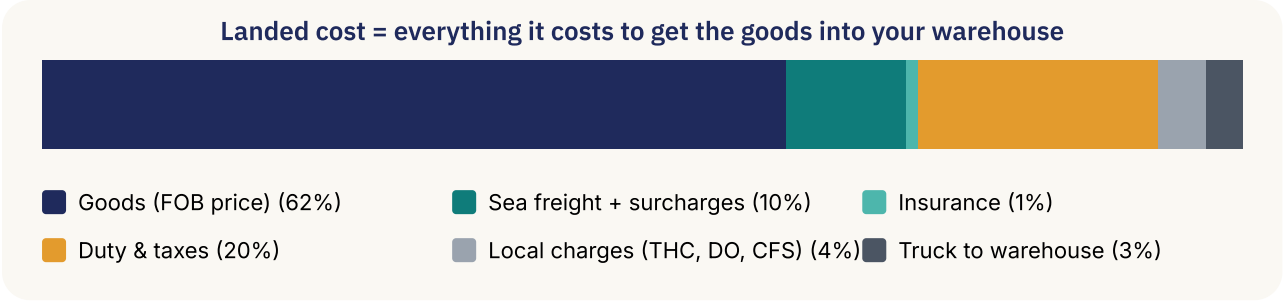
Freight rates

The **freight rate** is the price of the main trip. Sea FCL is priced **per container** (for example USD per 20GP). Sea LCL is priced **per CBM or per 1,000 kg, whichever is more** (W/M). Air is priced **per chargeable kilo**, often cheaper per kg for bigger weight "breaks" (+45 kg, +100 kg, +300 kg, +500 kg, +1,000 kg). Rates go up and down with supply and demand, so quotes have a **validity date**.

Surcharges: the extras

Code	Name	What it is for
BAF	Bunker adjustment factor	Covers changes in the price of ship fuel ("bunker")
THC	Terminal handling charge	Lifting the container on or off the ship at the terminal. OTHC at origin, DTHC at destination
CAF	Currency adjustment factor	Covers exchange-rate changes
PSS	Peak season surcharge	Extra when demand is high
LSS	Low sulphur surcharge	Cost of cleaner fuel required since 2020
GRI	General rate increase	An announced rise in the basic rate
DOC	Documentation fee	Issuing the BL or DO
Seal / ISPS	Seal fee / port security	The seal and port security costs
FSC / SSC	Fuel / security surcharge (air)	Air freight versions of fuel and security costs
CFS	CFS charges	Packing or unpacking LCL cargo at the CFS

Landed cost: the true price

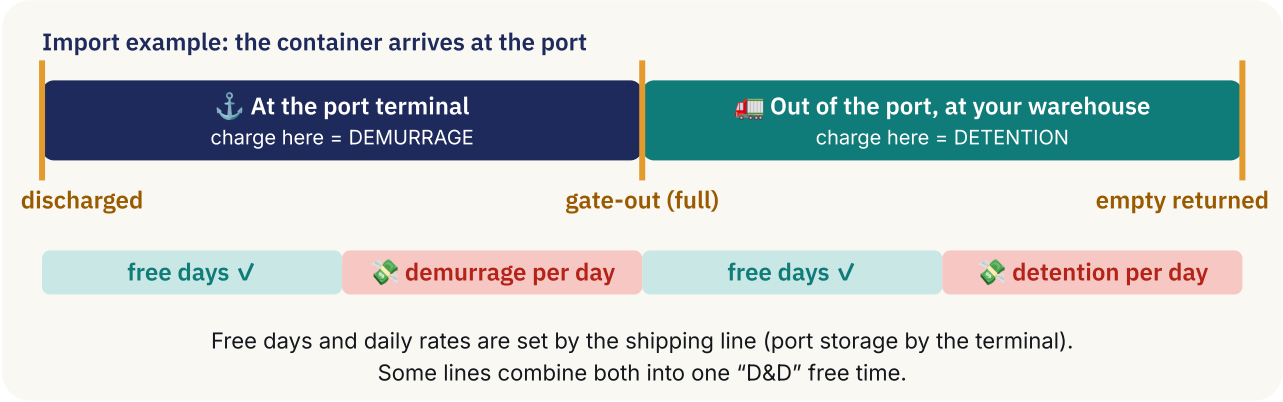


An illustration only. Real shares differ by product, route and duty rate.

Landed cost = price of the goods + freight + insurance + duty and taxes + all local charges + delivery. Two quotes can look very different once you add everything up. Ternary Cargo's "Get a price" page compares carriers including the full landed cost.

Demurrage and detention (D&D)

The shipping line lends you its container for free for a few days (**free time**). Keep it longer and you pay every day. The charges grow quickly and are one of the biggest surprise costs for importers.



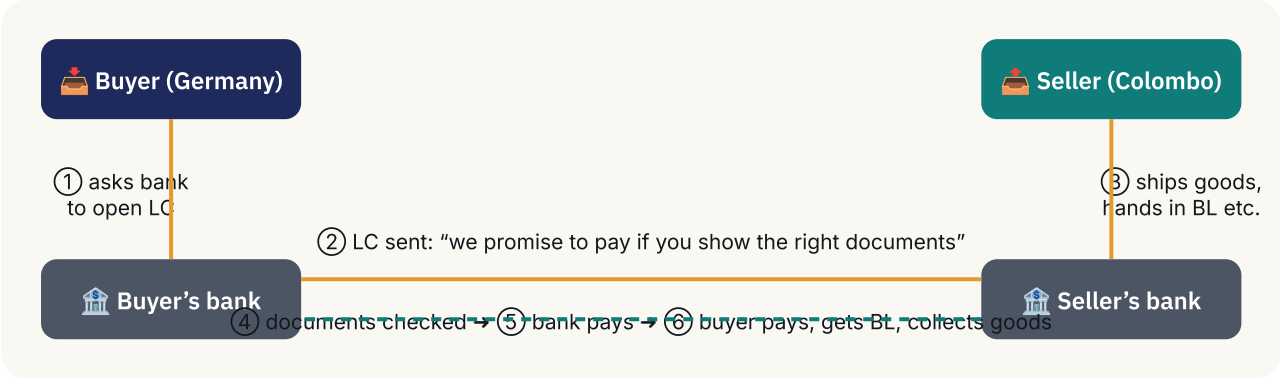
Demurrage = box still inside the port. Detention = box outside the port, not yet returned.

> 💡 **Think of it like this**
>
> Demurrage and detention are like late fees at a library. The library (shipping line) lends you the book (container) for free for a while. If you don't bring it back in time, you pay a fine for every extra day. Ternary Cargo shows a countdown for every container so you never forget.

How the seller gets paid

Method	How it works	Safer for
Cash in advance	Buyer pays before shipping	Seller
TT (telegraphic transfer)	A bank-to-bank electronic payment (wire). Can be in advance, part in advance, or after shipment	Depends on timing
LC (letter of credit)	The buyer's bank promises to pay when the seller presents exactly the documents listed in the LC. Usually follows the ICC rules called UCP 600	Both, if documents are perfect
D/P (documents against payment)	Banks hand the BL to the buyer only when the buyer pays	Fairly safe for seller

Method	How it works	Safer for
D/A (documents against acceptance)	Buyer gets documents by promising to pay later (signing a bill of exchange)	Buyer
Open account	Goods shipped, buyer pays later, e.g. in 60 days	Buyer



A letter of credit: banks stand in the middle so the seller is paid only if the documents are right, and the buyer gets the goods only after paying.

⚠️ Watch out!

With an LC, banks check **documents, not goods**. A tiny difference, such as a spelling, a weight or a late shipment date, can be a "discrepancy" and delay payment. Check every document against the LC before you ship.

Remember!

- FCL is priced per container; LCL per CBM or tonne (W/M); air per chargeable kg.
- **BAF** = fuel; **THC** = terminal handling; plus many other surcharges.
- **Landed cost** = goods + freight + insurance + duty/taxes + local charges + delivery.
- **Demurrage** = container overstays in the port. **Detention** = container kept outside too long. Watch the free days!
- **TT** = bank transfer. **LC** = bank promise to pay against correct documents.

CHAPTER 11

Tracking and arrival times

"When will it arrive?" is the question everyone asks. The honest answer is a range, not a single date.

In this chapter you will learn: how tracking works, the milestones, why ships and planes are late, and how to read a predicted ETA range like P50 and P90.

How tracking works

Each time something happens to your cargo, a **milestone** (event) is recorded: booked, gated in, loaded, departed, transhipped, arrived, discharged, customs released, gated out, delivered, empty returned. Shipping lines and airlines share these events. Ships also send their position by radio, called **AIS**, so a map can show where the ship is right now.

Short form	Means
ETD / ATD	Estimated / actual time of departure
ETA / ATA	Estimated / actual time of arrival
ETB	Estimated time of berthing (the ship tied up at the quay)
POL / POD	Port of loading / port of discharge
T/S	Transhipment

Why cargo is late



Port congestion

Too many ships waiting for a berth. Ships queue outside like cars at a busy junction.



Weather

Monsoon storms, rough seas, fog or cyclones slow ships and close ports.



Missed transhipment

The first ship is late, so the container misses its connecting ship and waits for the next one.



Rolled cargo

The ship is full or a cut-off was missed, so the container is left for a later sailing.



Blank sailings

The line cancels a sailing to save money when demand is low.



Route changes

For example, when ships avoided the Red Sea in 2024 and sailed around Africa, trips to Europe took longer.



Customs holds

Missing papers, wrong HS codes, or an inspection.



Paperwork problems

No original BL, unpaid charges or a wrong consignee name stops release.

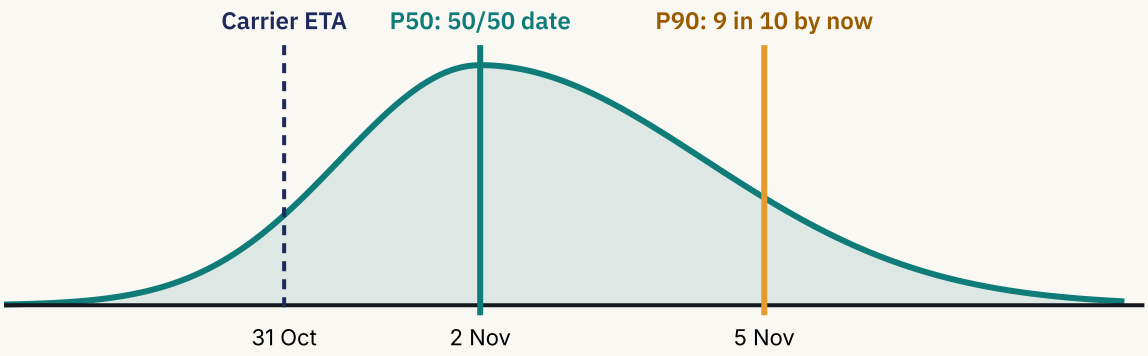


Inland delays

No trucks, no empty containers, strikes or traffic.

Predicted ETA: a range, not a promise

The carrier gives one ETA date. A **predicted ETA** looks at extra clues (congestion, weather, transhipment, how reliable this carrier usually is) and gives a range:



P50 = the date it is as likely to arrive before as after. P90 = we are 90% sure it will have arrived by then.

> 💡 **Think of it like this**
>
> If your friend says “I’ll be there at 4 o’clock”, that is the carrier ETA. If you know there is traffic, you think, “Probably 4:15, and almost certainly by 4:40.” 4:15 is your **P50** and 4:40 is your **P90**. Plan important things around the P90.

Ternary Cargo shows the carrier ETA, the P50 and P90, a confidence level and the delay factors (for example “Port congestion (Colombo) +1.2 days”). In the current prototype this is a simple, clearly labelled **“Prototype estimate”** built from mock data, not a trained model yet.

Remember!

- Tracking = a list of **milestones** plus, for ships, the **AIS** position.
- Common delays: congestion, weather, missed transhipment, rolled cargo, blank sailings, customs holds, missing papers.
- **ETA** is a guess. A good prediction is a range: **P50** (50/50) and **P90** (90% sure).
- Plan your warehouse and trucks around the P90, and keep the buyer updated.

CHAPTER 12

When things go wrong

Even with great planning, sometimes boxes get wet, cartons go missing or ships are late. What matters is spotting it fast and following the right steps.

In this chapter you will learn: the common problems, how to record them, and how to make a claim.

Problem	What it looks like	What to do first
💥 Damage	Crushed, torn, broken or leaking packages	Photograph before moving anything. Note it on the delivery receipt and tally sheet. Keep the packing.
💧 Wet cargo	Water stains, mould, rust. Often "container rain": condensation dripping from the roof	Photograph, check the container roof for holes, measure moisture. Use desiccants (moisture absorbers) next time.
➖ Shortage	Fewer packages than the packing list or ASN	Re-count, scan every SSCC, check the seal number, write "short landed" on the papers.
➕ Overage	More packages than expected	Do not take them into stock. Report them; they may belong to another shipment.
⚖️ Overweight	VGM or weighbridge weight higher than declared; over road limits	Stop and correct before loading. Repack or move cargo to another container.
⏰ Delay	Missed sailing, late arrival, held at customs	Tell the buyer early with a new ETA range; check D&D free days; ask for extra free time.
🔓 Seal problem	Broken, missing or different seal number	Photograph, record, call the carrier and insurer before opening.
🌡️ Temperature	Reefer too warm or too cold	Download the reefer's temperature log; keep samples; call the surveyor.
📄 Document error	Wrong name, weight, HS code or marks	Ask the issuer to correct it (BL corrections can cost a fee) before the goods arrive.

Tolerance: how close is close enough?

Scales and cartons are never perfect. A **tolerance** is the small difference you accept. Ternary Cargo flags any package whose weight is more than **±2%** away from the declared weight. Anything outside the tolerance becomes an **exception** to look at.

Making a claim

1

👀 Spot it and record it

Note the problem on the delivery receipt or EIR ("3 cartons crushed"), take photos and keep everything.

2

📣 Tell everyone quickly

Give written notice to the carrier and your insurer straight away. Time limits are short: for sea cargo under the widely used Hague-Visby Rules, notice is due at delivery, or within 3 days if the damage could not be seen; for air under the Montreal Convention, within 14 days for damage and 21 days for delay.

3

🔍 Survey

The insurer may send a **surveyor** to inspect and write a report.

4

📂 Send the claim

Include the invoice, packing list, BL or AWB, photos, tally sheet, survey report and your calculation of the loss.

5

💸 Get paid

Carriers' liability is limited by law and by the contract, often much less than the goods' value. That is why **cargo insurance** is so important.

⚓ General average: when everyone shares the loss

If a ship is in danger and the captain must throw some cargo overboard, or pay for a rescue, to save the ship, **everyone with cargo on board** shares the cost, even if their own goods arrived safely. Cargo insurance normally covers this. Without it, your cargo may be held until you pay a deposit.

⭐ Top tip

Most claims are lost because of **missing proof**. Scan every pallet, check every seal, photograph every problem and sign nothing as "in good order" until you have checked. Ternary Cargo keeps a timeline of scans, seals and photos for every hand-over.

Remember!

- Common problems: damage, wet cargo, shortage, overage, overweight, delays, seal problems, temperature, document errors.
- First: **photograph and write it down** before moving anything.
- Tell the carrier and insurer **in writing, quickly**. Time limits can be as short as 3 days.
- Carriers' liability is limited, so buy **cargo insurance**.
- Good proof (scans, seals, photos, tally sheets) wins claims.

CHAPTER 13

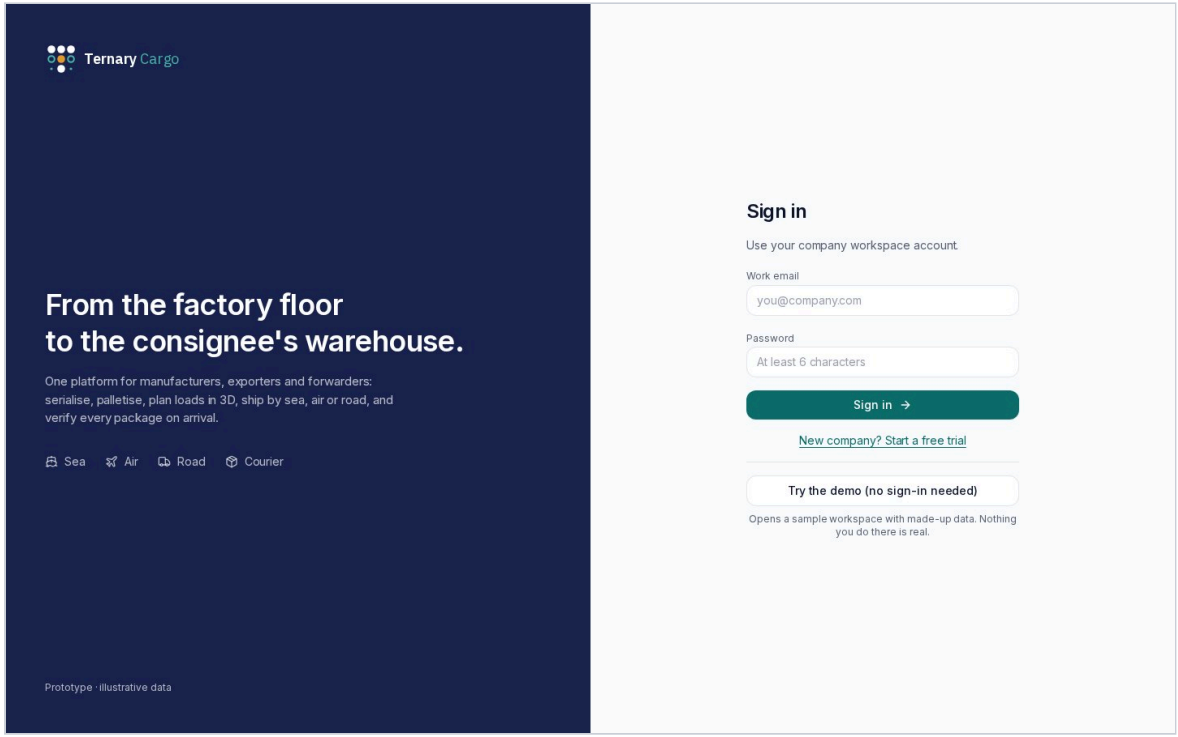
Using Ternary Cargo

Now you know how cargo works, let's use the system that keeps it all together. Every screen here is a real screenshot of the Ternary Cargo prototype.

In this chapter you will learn: how to sign in, find your way, follow a shipment, run the factory-to-container flow, plan a load, count cargo on arrival, share with a customer, and switch between Simple and Expert.

> ⚠️ **Before you start**
>
> This is the **prototype**. All companies, people, shipments and numbers in it are made-up demo data (Ceylon Freight Lines, Ceylon Tiles Mfg, Singer Sri Lanka PLC and so on). Predicted ETAs, rates and the AI assistant are mocked and labelled. Nothing you do in the demo is real, and data is kept only in your browser.

1. Signing in

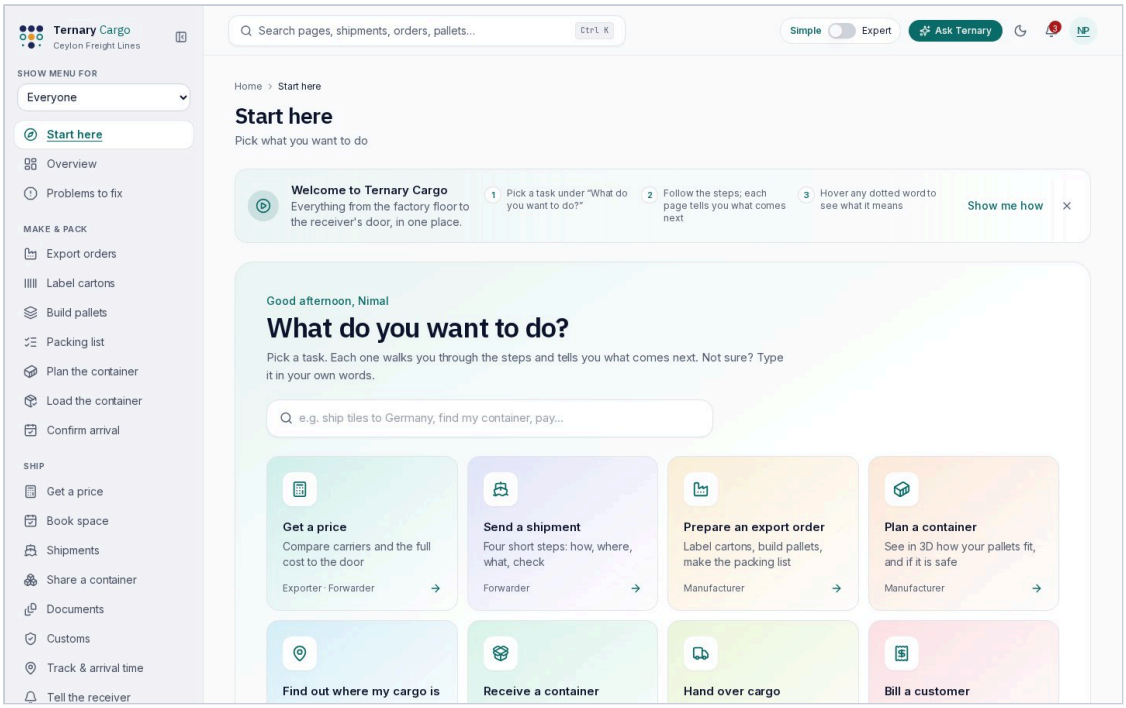


The sign-in page. Use your company workspace email and password, start a free trial, or press **Try the demo** to explore with sample data.

2. The Start page: "What do you want to do?"

After you sign in you land on **Start here**. Instead of a maze of menus, it asks a simple question and shows task tiles such as **Get a price**, **Send a shipment**, **Prepare an export order**, **Plan a container**, **Find**

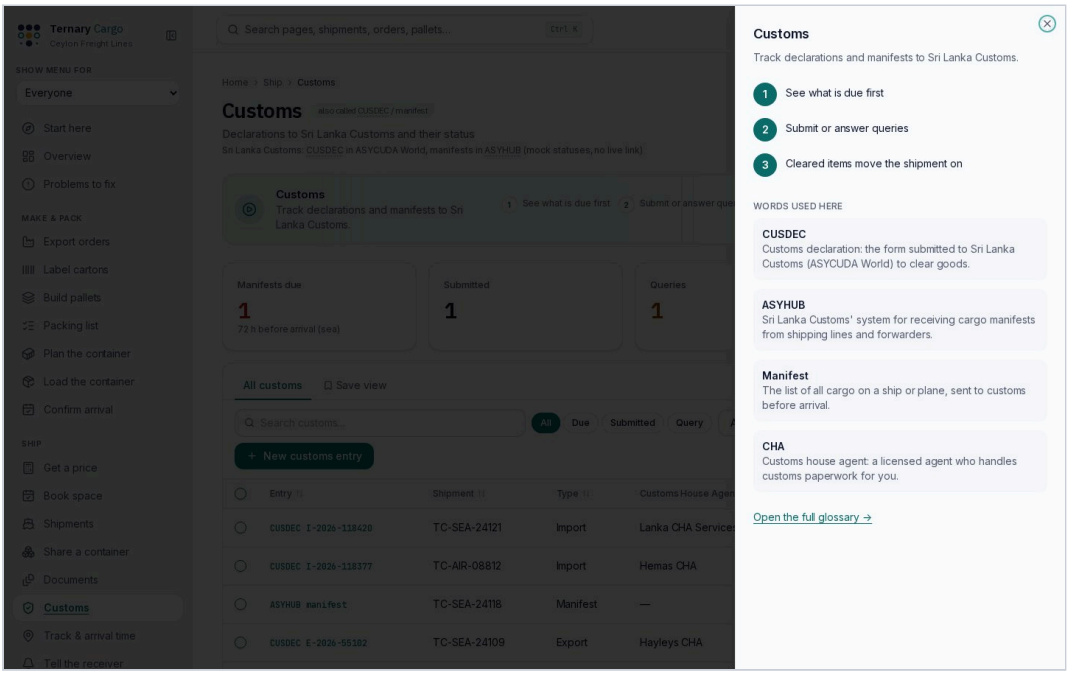
out where my cargo is, **Receive a container**, **Hand over cargo** and **Bill a customer**. Each tile walks you through the steps and tells you what comes next. You can also just type what you want in your own words.



Start here: pick a task tile. Below the tiles are the things that need you today, and journeys with their next step.

⭐ Help is everywhere

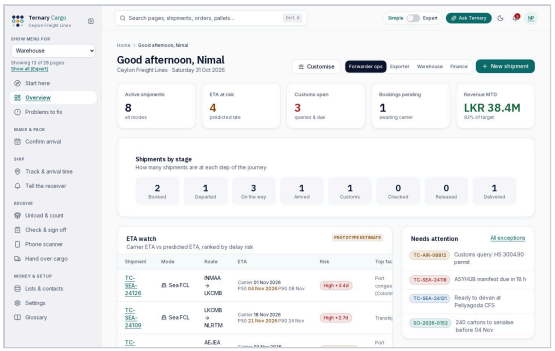
On most pages, a green **How this works** banner shows three numbered steps. **Show me how** opens a side sheet with the steps and the "Words used here". A word with a dotted underline shows its meaning when you hover over it.



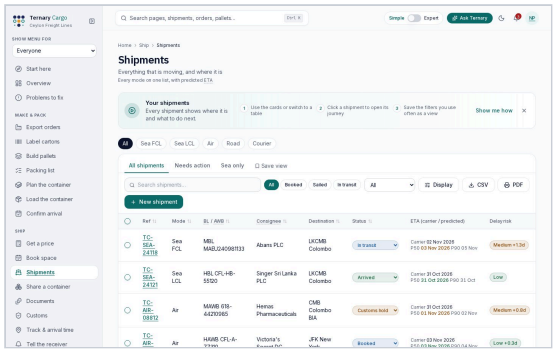
A per-page guide sheet: steps in plain words, and the cargo terms used on that page.

3. Finding your way: the menu

The left menu is grouped the way cargo flows: **Make & pack** (Export orders, Label cartons, Build pallets, Packing list, Plan the container, Load the container, Confirm arrival), **Ship** (Get a price, Book space, Shipments, Share a container, Documents, Customs, Track & arrival time, Tell the receiver), **Receive** (Unload & count, Check & sign off, Phone scanner, Hand over cargo) and **Money & setup**. At the top, **Show menu for** lets you pick a role (for example Warehouse) so you only see the pages that role needs. **Show all (Expert)** brings everything back. Press **Ctrl K** to search for any page, shipment, order or pallet.



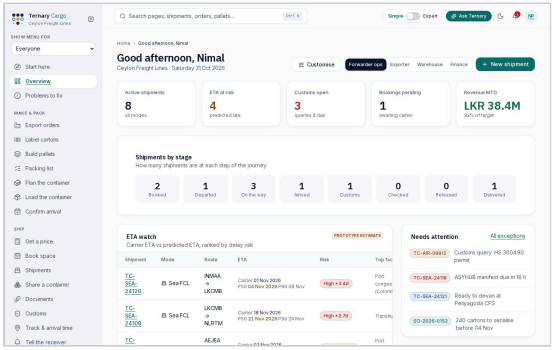
The menu for the Warehouse role: 13 of 28 pages.



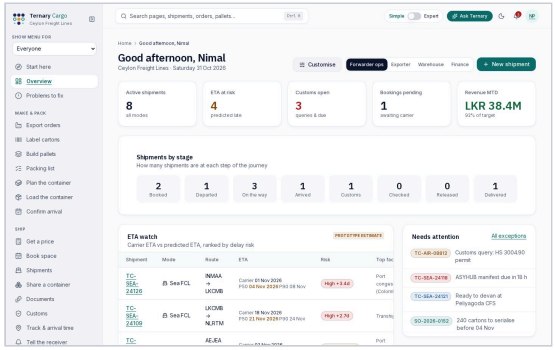
Shipments: every mode on one list, with status, carrier ETA and predicted ETA.

4. The Overview dashboard

Overview shows the numbers for your role: active shipments, ETA at risk, open customs items, pending bookings and revenue. **Shipments by stage** counts how many are at each step. **ETA watch** ranks shipments by delay risk, and **Needs attention** lists the problems to fix, such as “ASYHUB manifest due in 18 h”. Switch between Forwarder ops, Exporter, Warehouse and Finance views at the top. **Customise** lets you show, hide and reorder the cards.



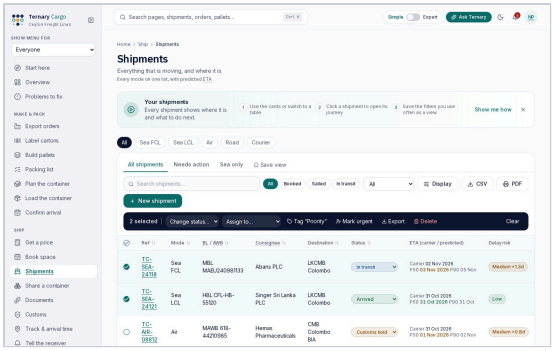
The Overview dashboard for forwarder operations.



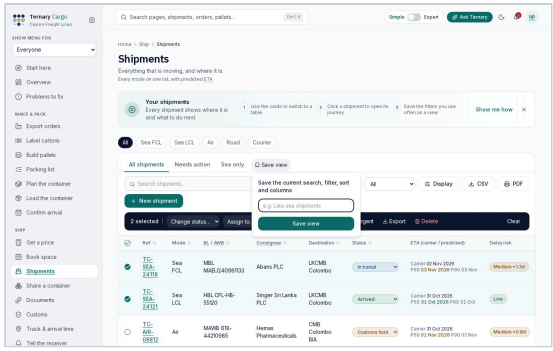
Customise: show, hide and reorder the widgets.

5. Lists, saved views and bulk actions

Every list (shipments, orders, pallets…) works the same way. Filter by mode or status, search, and choose columns under **Display**. **Save view** keeps your favourite filters under a name, like “Needs action”. Tick several rows to get a **bulk bar**: change status, assign, tag, mark urgent, export or delete them all at once. **CSV** and **PDF** export the list.



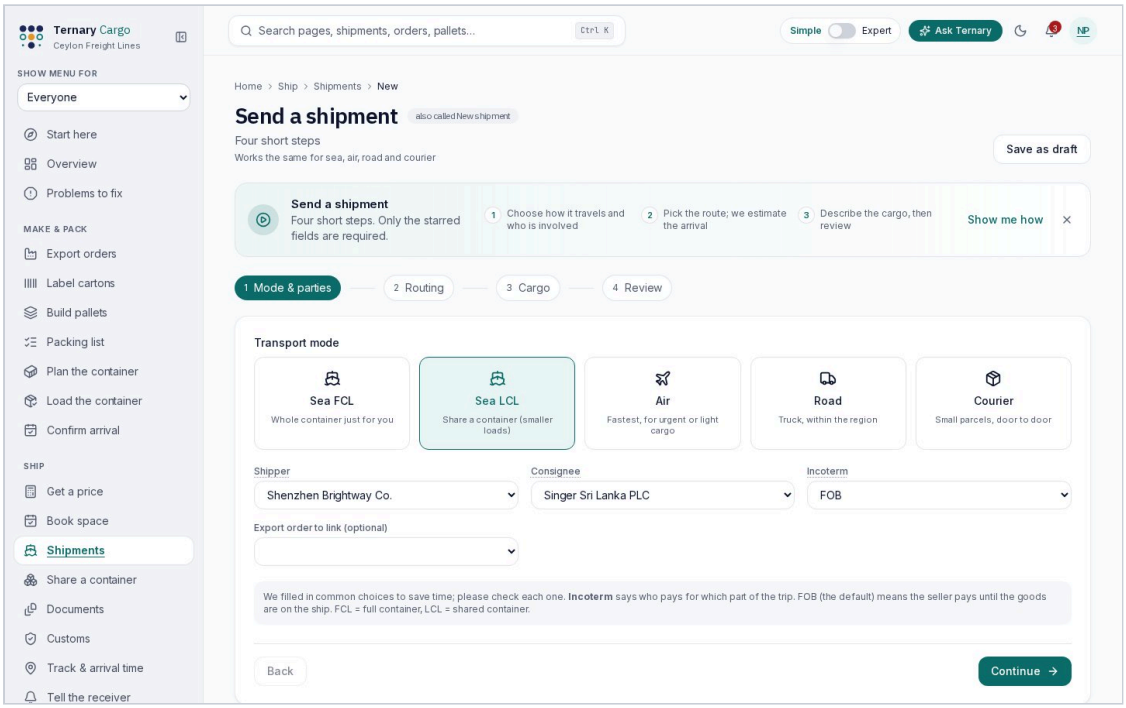
Tick rows to act on many at once.



Save the current filters as a named view.

6. Sending a shipment: the booking wizard

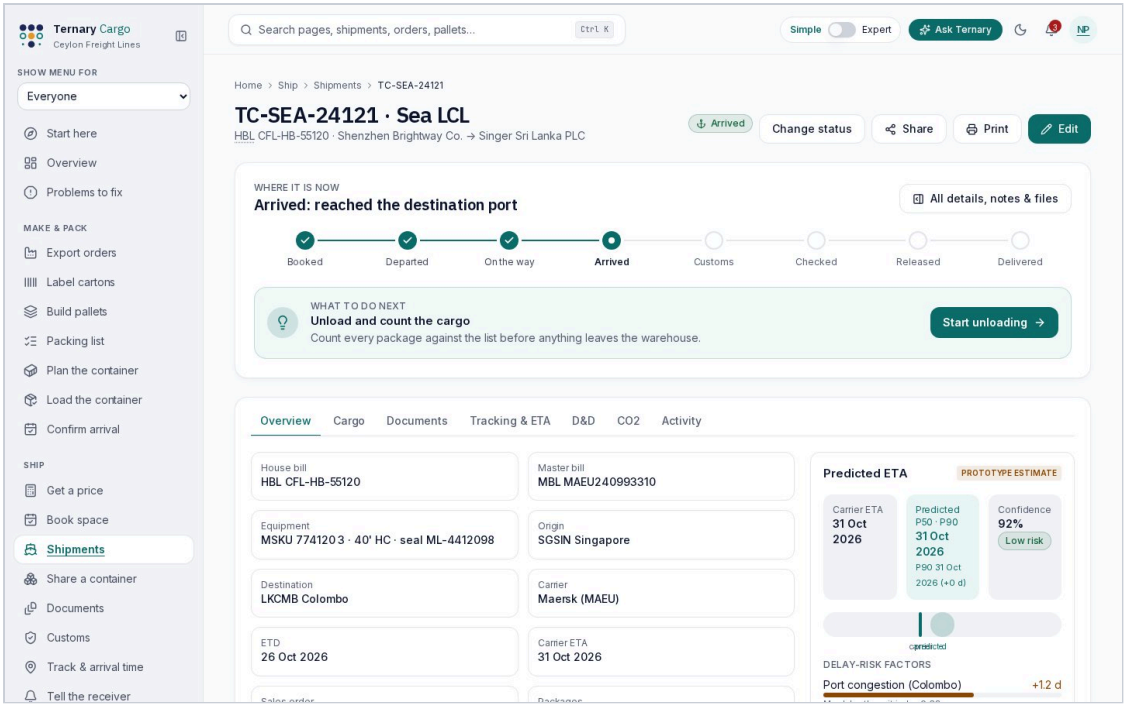
Send a shipment takes four short steps: **1 Mode & parties** (Sea FCL, Sea LCL, Air, Road or Courier; shipper, consignee and Incoterm), **2 Routing** (origin and destination; it estimates the arrival), **3 Cargo** and **4 Review**. Only the starred fields are required, and common choices are filled in for you to check.



Step 1 of the wizard. Each mode tile explains itself in plain words, e.g. “Sea LCL: share a container (smaller loads)”.

7. A shipment’s journey page

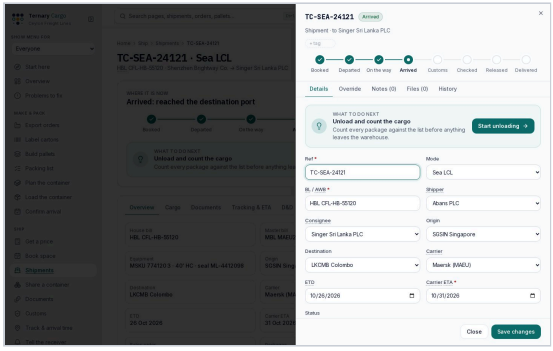
Open any shipment to see its **journey stepper**: Booked → Departed → On the way → Arrived → Customs → Checked → Released → Delivered. Under it, a **What to do next** card tells you the next job and gives you a button for it (here: **Start unloading**). Tabs show the **Overview** (house bill, master bill, container and seal, origin, destination, carrier, dates), **Cargo**, **Documents** (with a checklist of missing papers), **Tracking & ETA**, **D&D** (the free-time countdown), **CO2** and **Activity**.



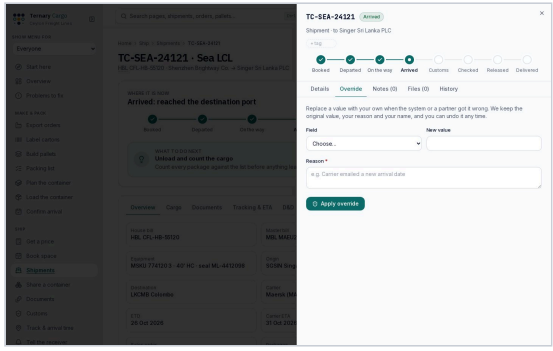
TC-SEA-24121, Sea LCL from Shenzhen to Colombo. It has arrived; the next step is to unload and count the cargo.

The record sheet: details, override, notes and files

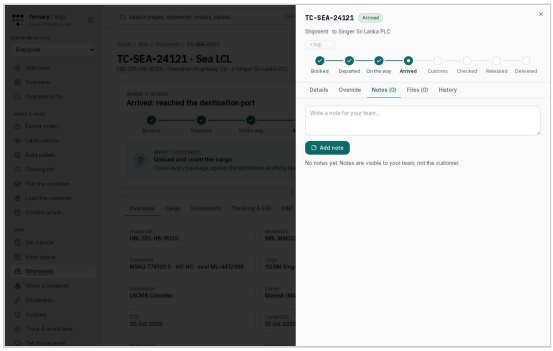
All details, notes & files opens a side sheet for any record. **Details** lets you edit every field (including your own custom fields). **Override** replaces a value when the system or a partner got it wrong. You must give a reason; the original value is kept and you can undo it. **Notes** and **Files** keep comments, attachments and photos on the record. **History** shows every change.



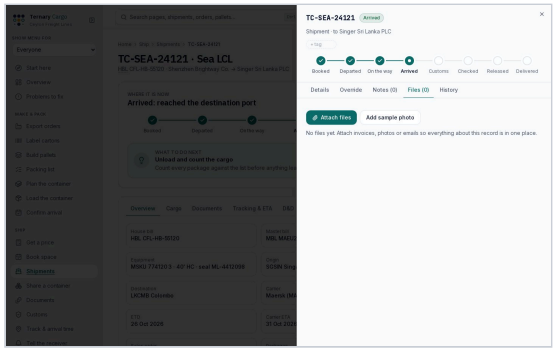
Details tab: edit fields and custom fields.



Override tab: a reason is required, and you can revert.



Notes on any record.



Attachments and photos.

8. The manufacturing flow: from order to container

This is the path a Sri Lankan factory follows. Each page tells you the next step.



1 🧾 **Export orders (sales orders)**

Every buyer order starts here: buyer, product, number of cartons and ship-by date, e.g. **SO-2026-0152**, 240 cartons of tiles for a buyer in Jebel Ali. Open it to see its journey and the next step.

2 🏷️ **Label cartons (serialisation)**

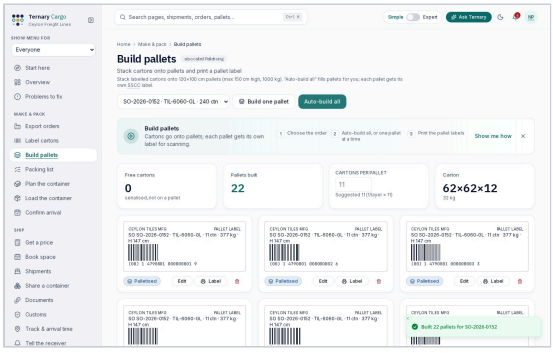
Choose the order and click **Generate labels**. Each carton gets a unique serial (SGTIN), so you can prove exactly what was shipped. Scan a few to check that the labels print and read.

3 🧱 **Build pallets (palletising)**

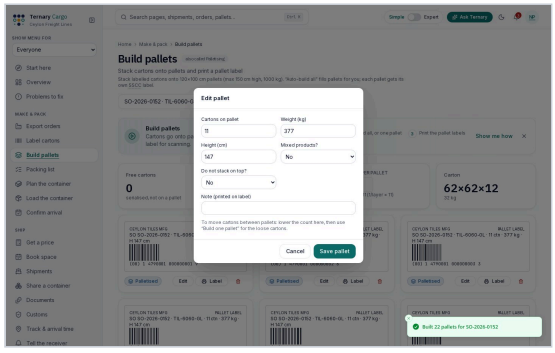
Choose the order and click **Auto-build all** (or build one pallet at a time). Cartons go onto 120 × 100 cm pallets (max 150 cm high, 1,000 kg), and each pallet gets its own **SSCC** label. Badges show **MIXED** or **DO NOT STACK**. **Edit** a pallet to change its cartons, weight, height, flags or add a note.

4 📋 **Packing list**

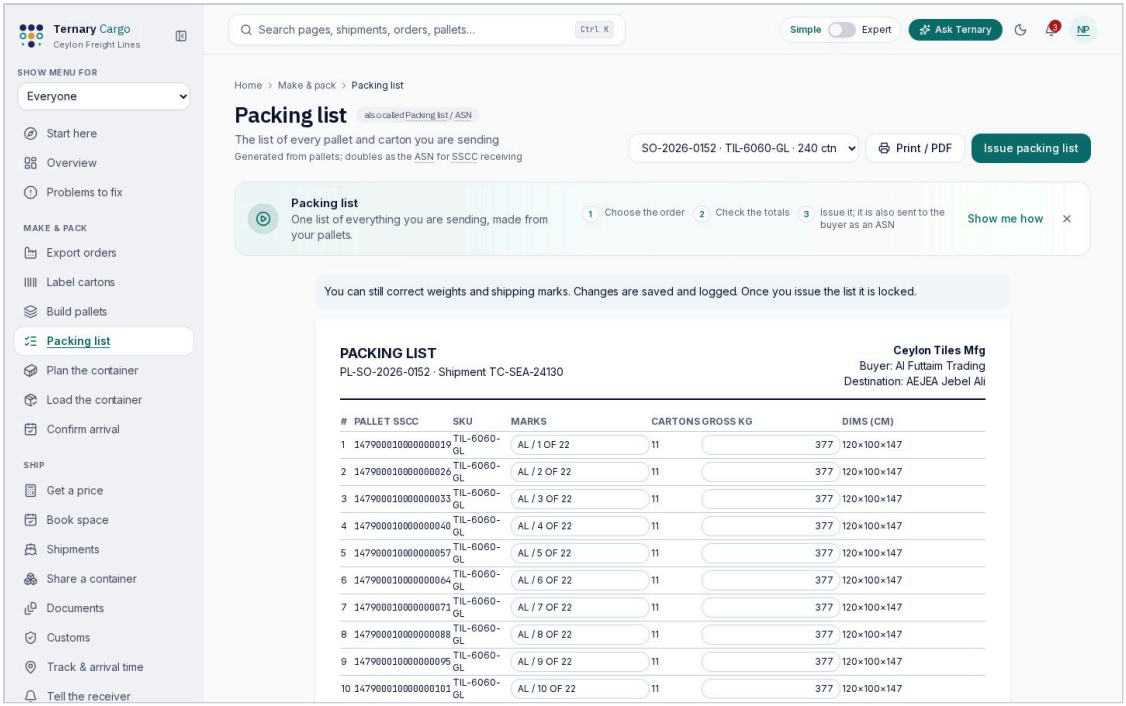
Made automatically from your pallets: every pallet's SSCC, SKU, marks, cartons, gross weight and size. You can still correct weights and shipping marks; changes are saved and logged. **Issue packing list** locks it and sends it to the buyer as the **ASN**.



Build pallets: 22 pallets built for SO-2026-0152, each with an SSCC barcode.



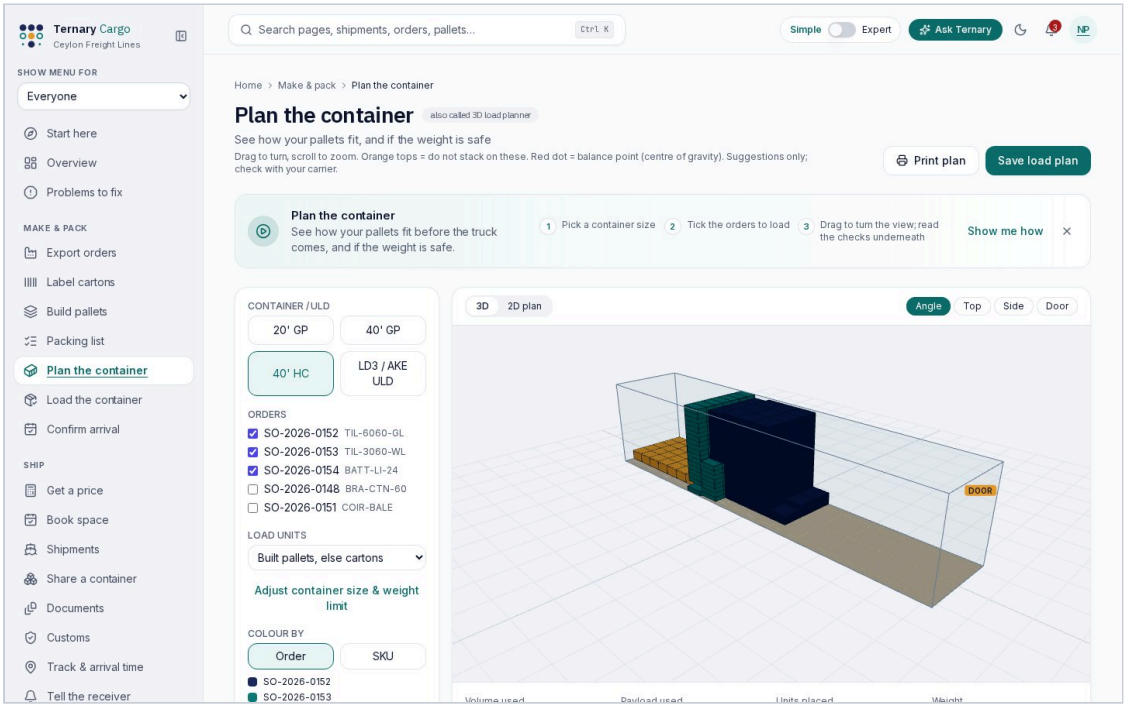
Edit a pallet: cartons, weight, height, flags and a note.



The packing list, made from the pallets. Editable until you issue it; then it is locked.

9. The load planner: “Plan the container”

Pick a container or ULD (**20′ GP**, **40′ GP**, **40′ HC** or **LD3/AKE**), tick the orders to load, and choose to load built pallets or cartons. The 3D view shows every unit coloured by order or SKU. Drag to turn it, scroll to zoom, or use the camera buttons **Angle, Top, Side, Door**. Orange tops mean “do not stack on these”. The red dot is the centre of gravity.

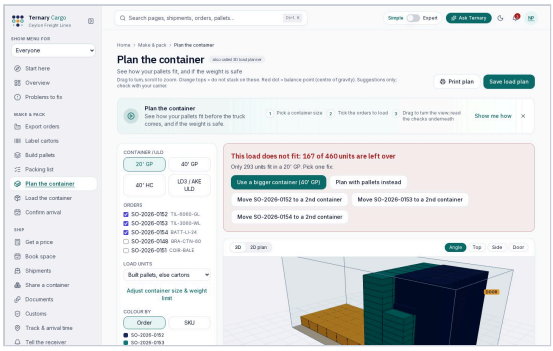


Three orders loaded into a 40′ HC. Underneath, the planner shows volume used, payload used, units placed and the weight.

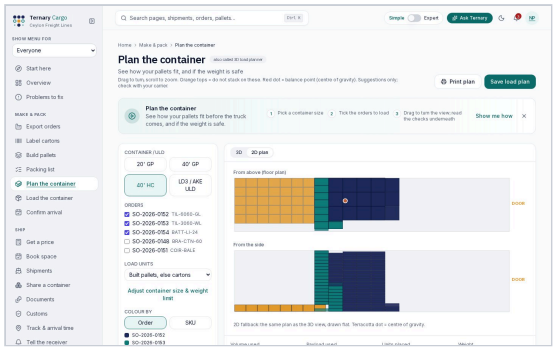
The checks underneath warn you if: the payload is over the limit (or above 95%); the centre of gravity is more than ±10% from the middle; the estimated road axle split is uneven; any **non-stackable** unit has cargo on top; or **dangerous goods** are on board (segregation and a DG declaration needed). These are

advice only: check with your carrier. The planner uses a simple, fast packing method, not a perfect optimiser.

When it does not fit, the planner is honest. A red banner says exactly how many units are left over and offers one-click fixes: use a bigger container, plan with pallets instead, or move an order to a second container. A **2D plan** (top and side views) is there if 3D is hard to read or print.



"This load does not fit: 167 of 460 units are left over" with fixes.

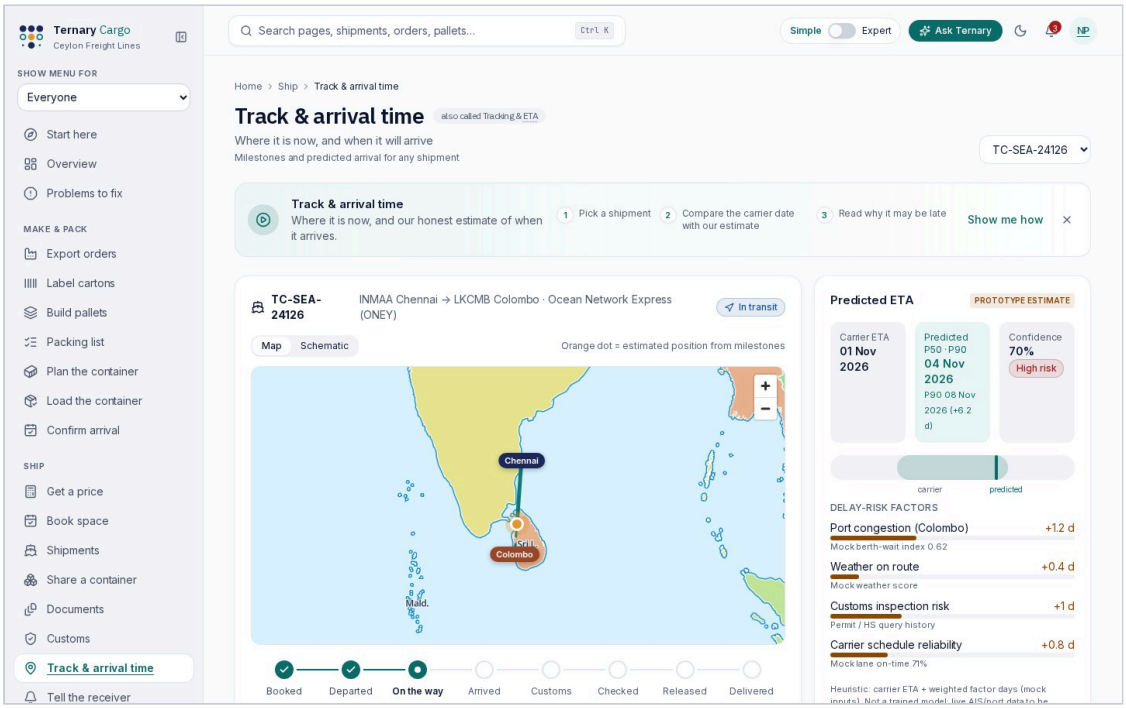


The 2D top and side plan.

Load the container comes next: scan each pallet as it goes in, type the container and seal numbers, and confirm loading. That scan record is your proof of what went in.

10. Tracking and arrival time

Track & arrival time shows the route on a map, the journey milestones, and the **Predicted ETA** card: carrier ETA, P50 and P90 dates, a confidence level, and the **delay-risk factors** such as port congestion, weather, customs inspection risk and carrier reliability. It is labelled **Prototype estimate**.



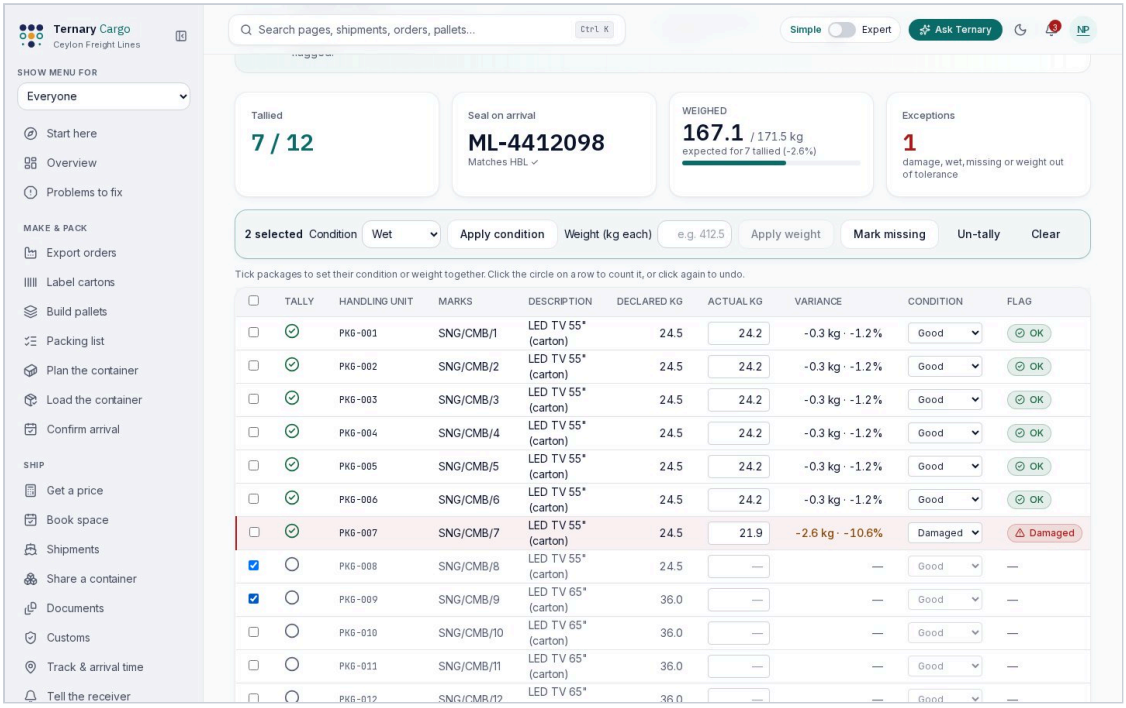
TC-SEA-24126, Chennai to Colombo on Ocean Network Express. Carrier ETA 1 Nov; predicted P50 4 Nov, P90 8 Nov; high risk.

11. Receiving: unload, count, check and release

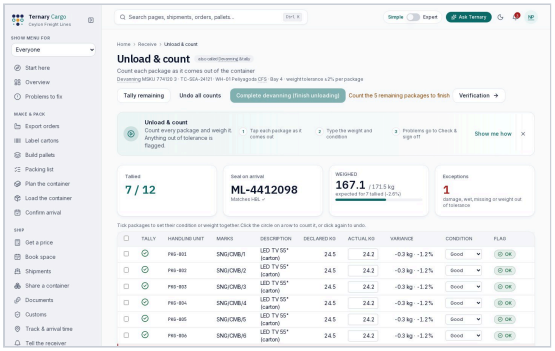
Unload & count (devanning and tally) lists every package expected in the container, here 12 cartons of LED TVs. Tick packages as they come out, type the actual weight and set the condition (Good,

Damaged, Wet…). The page shows how many are tallied, whether the **seal on arrival** matches the HBL, the total weighed against the expected weight, and the number of exceptions. Any package more than **±2%** away from its declared weight is flagged.

To save time, tick several packages and set their condition or weight together, **Mark missing**, or **Un-tally** them. **Tally remaining** counts the rest as Good for you to check. **Complete devanning** stays greyed out until every package is counted, and it tells you why (“Count the 5 remaining packages to finish”).



Two packages ticked for a bulk action. PKG-007 is flagged Damaged: 21.9 kg against 24.5 kg declared (−10.6%).



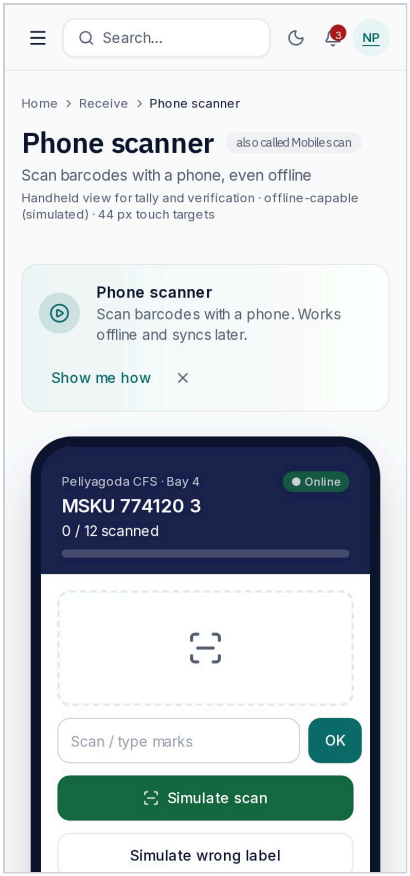
Complete devanning explains why it is not ready yet.

Check & sign off (verification)

Compare what arrived with the HBL and the packing list (ASN). Record damage with photos in a **discrepancy report** (e.g. DR-2026-0091). A supervisor signature is required. If a discrepancy is still open, you must save the report, give a reason and get a **second approver**: “Verified with exception”.

Hand over cargo (delivery order)

A delivery order can only be released when three gates are green: **customs cleared**, **verification signed off** and **charges paid**.



The phone scanner: works offline and syncs later.

The phone scanner

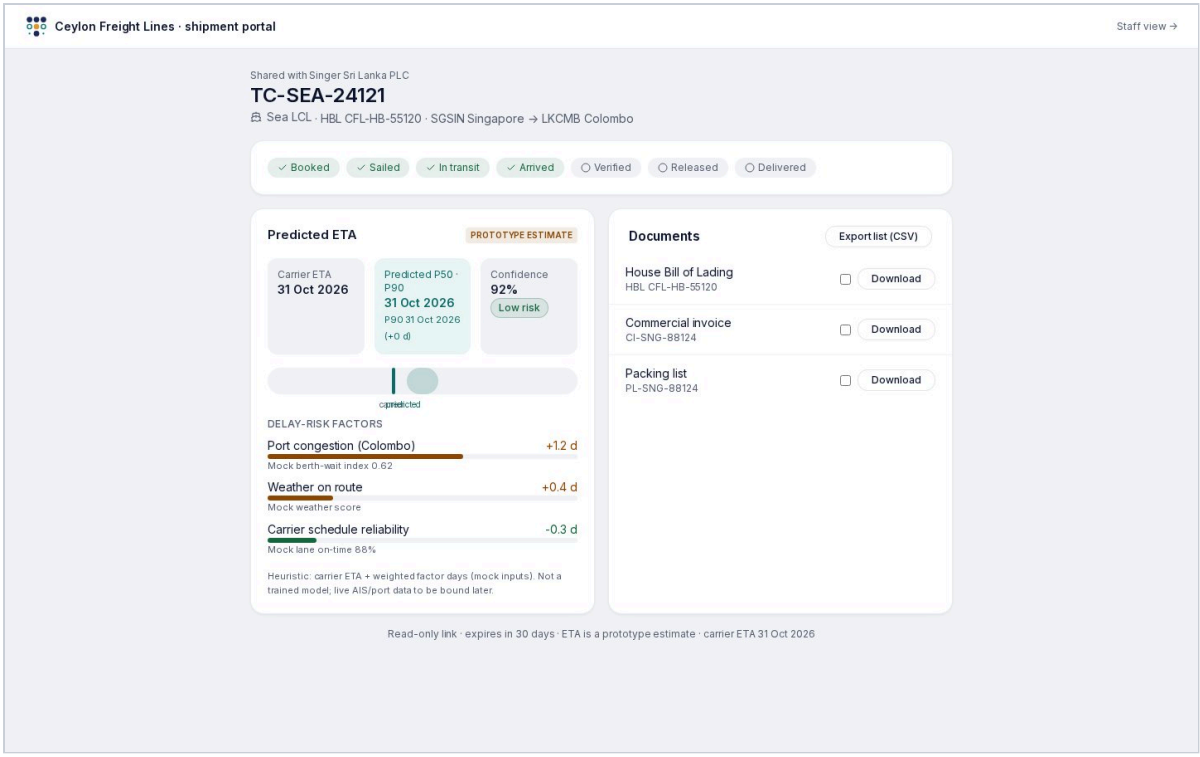
Warehouse staff can use a phone in the shed. Scan a barcode (or type the marks). Green means it was expected; red means it is not on the list. It keeps working without internet and syncs when you are back online. Buttons are big (44-pixel touch targets), so they are easy to tap.

Confirm arrival

For the factory flow, the buyer scans each pallet's SSCC on arrival. Missing pallets are flagged against the ASN before receipt is confirmed.

12. The customer portal

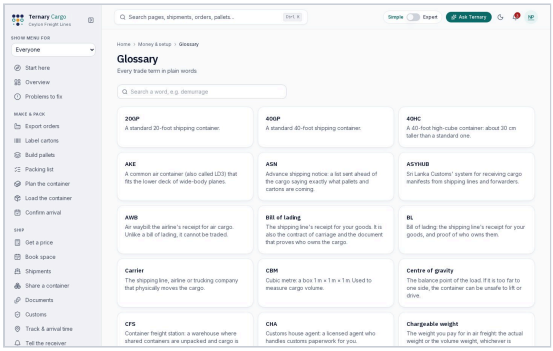
Share a read-only link with your customer (the consignee) so they can check their shipment without calling you. They see the journey, the predicted ETA with its reasons, and the **documents** (house BL, commercial invoice, packing list) to download as PDFs, or export the list as CSV. The link expires after 30 days.



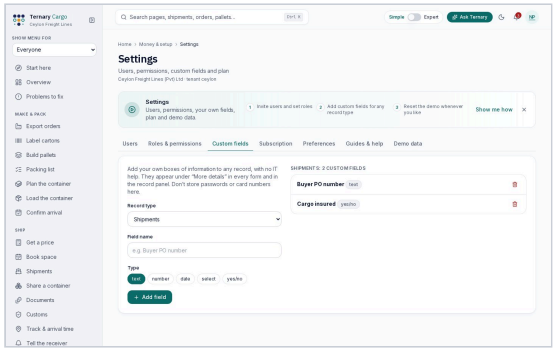
The portal page shared with Singer Sri Lanka PLC for TC-SEA-24121.

13. The glossary and your own fields

Glossary explains about 55 cargo terms in plain words, with a search box: type “demurrage” and it appears. **Settings → Custom fields** lets you add your own fields to any record type: text, number, date, select (a list of choices) or yes/no.



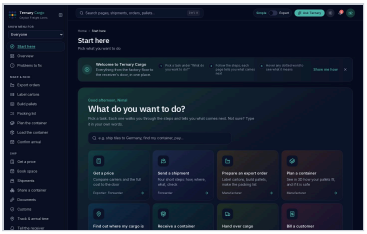
The in-app glossary: every trade term in plain words.



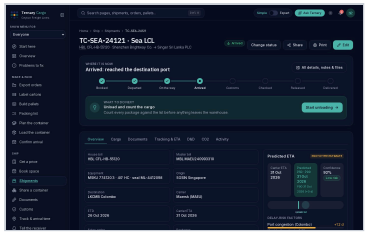
Add your own fields to any record.

14. Simple or Expert? Light or dark?

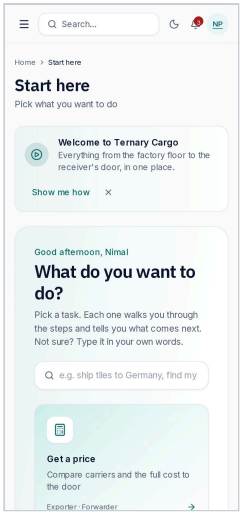
At the top of every page is a **Simple ⇄ Expert** switch. **Simple** (the default) hides advanced columns and settings so the screen is calm; you can double-click a cell to edit it. **Expert** shows every column and every page in the menu. **Nothing is ever removed**: switching only hides or shows. The moon button switches to **dark mode**, and the whole app also works on a phone.



Dark mode: Start page.



Dark mode: shipment journey.



On a phone.

Quick map: which page for which job?

I want to...	Go to
Get a freight price and landed cost	Ship → Get a price
Book a shipment	Ship → Shipments → New shipment (or Book space)
Prepare an export order	Make & pack → Export orders → Label cartons → Build pallets → Packing list
See if my cargo fits	Make & pack → Plan the container
Prove what I loaded	Make & pack → Load the container (scan + seal)
Lodge customs or a manifest	Ship → Customs
Know when it will arrive	Ship → Track & arrival time
Tell the receiver it's coming	Ship → Tell the receiver
Unload and count a container	Receive → Unload & count
Report damage and sign off	Receive → Check & sign off
Release the cargo	Receive → Hand over cargo
See what needs fixing	Problems to fix
Look up a word	Money & setup → Glossary

Remember!

- Start at **Start here** and pick a task; every page says what comes next.
- **Show me how** and dotted words explain everything; the **Glossary** has the rest.
- Factory flow: **order → label cartons → build pallets → packing list/ASN → plan container → load & seal → confirm arrival**.
- The load planner tells the truth: red banner = it does not fit, with fixes.
- On arrival: **unload & count → check & sign off → hand over** (customs + checks + payment).
- **Simple** hides advanced things; **Expert** shows everything. Nothing is removed.

CHAPTER 14

The A–Z glossary and cheat sheets

Every cargo word in this booklet, in plain words, plus one-page cheat sheets you can keep next to your desk.

In this chapter you will learn: where to look up any term fast, and the key numbers and rules on a few quick-reference pages.

A

Advance Shipping Notice (ASN): An electronic list sent to the buyer before goods arrive, saying exactly which pallets and cartons are coming.

Air waybill (AWB): The airline's receipt and contract for air cargo. Not proof of ownership. 11-digit number; first 3 digits = airline.

AIS: Automatic Identification System: the radio signal ships send with their position.

AKE / LD3: The most common air container; fits the lower deck of wide-body planes. Max gross 1,588 kg.

Arrival notice: A message telling the consignee when cargo arrives, the charges and the free time.

ASYCUDA World: Sri Lanka Customs' computer system for customs declarations (CUSDEC).

ASYHUB: The platform through which cargo manifests must be lodged with Sri Lanka Customs (since 7 May 2025).

ATA / ATD: Actual time of arrival / departure.

B

BAF: Bunker adjustment factor: a surcharge for changes in ship fuel prices.

Berth: The place at a quay where a ship ties up.

BIC code: The registered 3-letter owner code at the start of a container number.

Bill of lading (BL, B/L): The shipping line's receipt for goods, evidence of the contract of carriage, and (if negotiable) proof of ownership.

Blank sailing: A cancelled sailing.

Bonded warehouse: A warehouse where imported goods can be stored before duty is paid.

Booking: Reserving space on a ship, plane or truck.

Broker: See Customs broker.

C

CAF: Currency adjustment factor: a surcharge for exchange-rate changes.

Carrier: The company that actually moves the goods: shipping line, airline, trucker or railway.

CBM: Cubic metre (1 m × 1 m × 1 m). Used to measure cargo volume.

Centre of gravity (CoG): The balance point of a load. Keep it near the middle.

Certificate of origin (CO): A document proving where goods were made.

CFS: Container freight station: a warehouse where LCL cargo is packed into and unpacked from shared containers.

CHA: Customs House Agent: a licensed agent who lodges customs declarations in Sri Lanka.

Chargeable weight: What you pay for in air freight: the higher of actual weight and volumetric weight.

Commercial invoice: The seller's bill for the goods; used by customs.

Consignee: The person or company receiving the goods.

Consignor: Another word for shipper.

Consolidation: Combining several small shipments into one container or air pallet.

Container: A standard steel box for cargo: 20′, 40′, 40′ HC and special types.

CTU Code: The international code of practice for packing cargo transport units (containers, trucks).

CUSDEC: Sri Lanka's customs declaration form, lodged in ASYCUDA World.

Customs: The government agency that controls goods crossing the border and collects duty and taxes.

Customs broker: A licensed agent who handles customs formalities (CHA in Sri Lanka).

Cut-off: The deadline for cargo, documents, VGM or customs before a sailing or flight.

CY: Container yard: the area where containers are stored at a terminal.

D

Dangerous goods (DG, hazmat): Goods that can harm people, property or the environment. Governed by IMDG (sea) and IATA DGR (air).

Delivery order (DO): Permission to collect cargo from the terminal, CFS or carrier.

Demurrage: A charge when a full container stays at the port terminal beyond the free time.

Detention: A charge when a container is kept outside the terminal beyond the free time.

Devanning: Unloading cargo from a container (also destuffing or unstuffing).

Discrepancy: A difference between what was expected and what arrived.

Door-to-door: A service from the seller's premises to the buyer's premises.

Dunnage: Wood, bags or other material used to fill gaps and protect cargo.

Duty: A tax on imported (sometimes exported) goods.

E

EIR: Equipment interchange receipt: a record of a container's condition when it changes hands at a gate.

ETA / ETD / ETB: Estimated time of arrival / departure / berthing.

Exception: Anything that needs a person to look at it: damage, shortage, delay, missing document.

Export: Goods sent out of a country.

Exporter: The seller who sends goods abroad.

F

FCL: Full container load: one shipper's goods in one whole container.

FEU: Forty-foot equivalent unit (= 2 TEU).

Flat rack: A container with no roof or side walls, for oversized cargo.

Free time: The days you can use a container before demurrage or detention begin.

Freight: Cargo, or the charge for carrying it.

Freight forwarder: A company that organises shipments for others: bookings, documents, customs and delivery.

G

General average: A sea-law rule where all cargo owners share the cost of a sacrifice made to save the ship.

Gross weight: Total weight including packaging (and, for a container, the box).

GS1: The organisation that issues barcode numbers (GTIN, SSCC). Sri Lanka's prefix is 479.

GS1-128: A barcode type used on cartons and pallets, with Application Identifiers in brackets.

GTIN: Global Trade Item Number: identifies a type of product.

H

Haulier: The company or driver that moves goods by road.

HAWB: House air waybill: issued by a forwarder to its shipper.

HBL: House bill of lading: issued by a forwarder to its shipper.

High cube (HC): A container about 30 cm taller than standard (9′6″ high).

HS code: Harmonized System code: the international number used to classify goods for customs.

I

IATA DGR: The air rules for dangerous goods.

ICC: International Chamber of Commerce: writes Incoterms and UCP 600.

IMDG Code: The sea rules for dangerous goods.

Import: Goods brought into a country.

Importer: The buyer who brings goods in.

Incoterms®: ICC rules for who pays, who carries the risk and who does what between seller and buyer. Current version: 2020.

Insurance certificate: Proof that cargo is insured.

ISO 6346: The standard for container numbers.

ISPM 15: The international standard for treating wooden packaging; look for the stamp.

L

Landed cost: The total cost of getting goods into your warehouse.

Lashing: Straps, chains or ropes that tie cargo down.

LC (letter of credit): A bank's promise to pay the seller if the right documents are presented.

LCL: Less than container load: sharing a container with other shippers.

LKCMB: The UN/LOCODE for Colombo port.

Load plan: A plan showing where every unit goes in a container or truck.

M

Manifest: The list of all cargo on a ship, plane or truck, lodged with customs.

MAWB: Master air waybill: issued by the airline to the forwarder.

MBL: Master bill of lading: issued by the shipping line to the forwarder.

Milestone: A tracked event in a shipment's journey.

Multimodal: Using more than one mode of transport under one contract.

N

Net weight: Weight of the goods without packaging.

Notify party: The person to be told when cargo arrives (often the buyer or its broker).

NVOCC: Non-vessel operating common carrier: issues its own bills of lading but does not own ships.

O

Open top: A container with a removable tarpaulin roof for tall cargo.

Origin: Where a shipment starts.

Out of gauge (OOG): Cargo bigger than a standard container.

Overage: More packages arrived than expected.

P

P50 / P90: Predicted ETA dates: 50% likely to arrive before P50; 90% likely to have arrived by P90.

Packing list: A list of everything in a shipment, by package, with weights and sizes, but no prices.

PAL: Port and Airport Development Levy (Sri Lanka).

Pallet: A flat platform for stacking goods so a forklift can move them.

Palletising: Stacking and securing cartons on pallets.

Payload: The most cargo a container, truck or ULD can carry.

PMC / PAG: Common air cargo pallets (PMC 96 × 125 in, PAG 88 × 125 in).

POD: Port of discharge, or proof of delivery (depends on context).

POL: Port of loading.

Proforma invoice: A preview invoice sent before the sale is final.

R

Reefer: A refrigerated container.

Release: Permission for goods to leave customs or the terminal.

Rolled: Cargo left behind for a later sailing or flight.

S

Sea waybill: A non-negotiable sea transport document; the named consignee collects the goods.

Seal: A numbered lock on container doors; must match the BL.

SGTIN: Serialised GTIN: a product number plus a unique serial, for one exact item.

Shipper: The person or company sending the goods.

Shipping instructions (SI): Details given to the carrier to produce the BL.

Shipping marks: Words and numbers on cartons that match them to the documents.

Shortage: Fewer packages arrived than expected.

SOLAS: The international convention for the Safety of Life at Sea, which includes the VGM rule.

SSCC: Serial Shipping Container Code: an 18-digit GS1 number for one logistic unit (pallet, parcel).

SSCL: Social Security Contribution Levy (Sri Lanka).

Stuffing: Packing cargo into a container.

Surcharge: An extra charge added to the basic freight rate.

Surveyor: An independent inspector who reports on damage or condition.

T

Tally: Counting and checking packages during loading or unloading.

Tare: The weight of an empty container or package.

Telex release: Releasing cargo at destination without the original BL, after it was surrendered at origin.

Terminal: The part of a port (or airport) where cargo is handled.

TEU: Twenty-foot equivalent unit: the size of one 20′ container.

THC: Terminal handling charge.

Tolerance: The small difference you accept, e.g. ±2% on weight.

Transhipment: Moving cargo from one ship (or plane) to another on the way.

Transit time: How long the trip takes.

TT: Telegraphic transfer: an electronic bank payment.

U

UCP 600: ICC rules for letters of credit.

ULD: Unit load device: an air container or pallet.

UN number: A four-digit number that identifies a dangerous substance, e.g. UN3480.

UN/LOCODE: Five-letter codes for places, e.g. LKCMB for Colombo.

V

VAT: Value added tax; 18% standard rate in Sri Lanka since 1 January 2024.

Verification: Checking that what arrived matches the documents, then signing off.

Vessel: A ship.

VGM: Verified gross mass: the confirmed weight of a packed container, needed before loading.

Volumetric weight: Air freight size-weight: L × W × H (cm) ÷ 6,000.

Voyage: One trip of a ship, with a voyage number.

W/M: Weight or measure: LCL is charged on whichever is higher, tonnes or CBM.

Warehouse: A building where goods are stored.

Cheat sheet 1: containers and ULDs

Unit	Inside L × W × H	Volume	Max payload (typical)	TEU
20GP	5.90 × 2.35 × 2.39 m	~33 m³	~28,000 kg	1
40GP	12.03 × 2.35 × 2.39 m	~67 m³	~26,500–28,800 kg	2
40HC	12.03 × 2.35 × 2.70 m	~76 m³	~28,600 kg	2
20′ reefer	~5.45 × 2.28 × 2.16 m	~28 m³	~27,000–29,000 kg	1
40′ HC reefer	~11.6 × 2.29 × 2.43 m	~67 m³	~29,000 kg	2
AKE / LD3	Base 156 × 153 cm, H 163 cm	~4.3 m³	1,588 kg gross	–

Typical values from carrier and ULD-owner published specifications. Always check the CSC plate and door markings of the actual unit.

Quick maths

- **CBM** = L × W × H in metres. 120 × 100 × 150 cm pallet = 1.8 m³.
- **Air volumetric kg** = L × W × H (cm) ÷ 6,000.
- **Chargeable weight** = the bigger of actual and volumetric.
- **LCL W/M**: 1 CBM is compared with 1,000 kg.
- **Gross** = tare + payload.
- **Weight tolerance** in Ternary Cargo: ±2% per package.
- **CoG check** in the planner: within ±10% of the middle.

Codes you will see

- **LKCMB** Colombo port · **CMB** Bandaranaike International Airport
- **DEHAM** Hamburg · **NLRTM** Rotterdam · **GBFXT** Felixstowe
- **AEJEA** Jebel Ali · **SGSIN** Singapore · **INMAA** Chennai
- Container no. = 4 letters + 6 digits + check digit
- SSCC = (00) + 18 digits · GTIN = (01) + 14 digits
- Sri Lanka GS1 prefix: **479**

Cheat sheet 2: Incoterms® 2020 on one page

Rule	Mode	Seller pays main freight?	Seller insures?	Risk passes to buyer…	Import duty
EXW	Any	No	No	At seller's premises, not loaded	Buyer
FCA	Any	No	No	When handed to buyer's carrier at named place	Buyer
CPT	Any	Yes	No	When handed to the first carrier	Buyer
CIP	Any	Yes	Yes, ICC (A)	When handed to the first carrier	Buyer
DAP	Any	Yes	No (but seller has the risk)	At named destination, ready for unloading	Buyer
DPU	Any	Yes	No (but seller has the risk)	At named destination, after unloading	Buyer
DDP	Any	Yes	No (but seller has the risk)	At named destination, ready for unloading	**Seller**

Rule	Mode	Seller pays main freight?	Seller insures?	Risk passes to buyer…	Import duty
FAS	Sea	No	No	Alongside the ship at port of loading	Buyer
FOB	Sea	No	No	On board the ship at port of loading	Buyer
CFR	Sea	Yes	No	On board the ship at port of loading	Buyer
CIF	Sea	Yes	Yes, ICC (C)	On board the ship at port of loading	Buyer

Export clearance is the seller's job in every rule except EXW. Use FCA/CPT/CIP rather than FOB/CFR/CIF for containers.

Cheat sheet 3: documents by mode

Document	Sea FCL	Sea LCL	Air	Courier
Commercial invoice	✔	✔	✔	✔
Packing list	✔	✔	✔	usually
Export CUSDEC (Sri Lanka)	✔	✔	✔	courier handles
Booking confirmation	✔	✔	✔	–
Shipping instructions	✔	✔	AWB details	–
VGM	✔	forwarder/CFS	–	–
Bill of lading	MBL (+HBL if forwarder)	HBL + MBL	–	–
Air waybill	–	–	MAWB + HAWB	courier waybill
Certificate of origin	if needed	if needed	if needed	if needed
Insurance certificate	CIF/CIP or own cover	same	same	optional
DG declaration	if DG	if DG	if DG	if DG (often refused)
Arrival notice / DO	✔	✔	✔	–

The journey in 14 words

Quote · Booking · Pickup · Stuffing · Export customs · Gate-in · Loading · Departure · Transit · Arrival · Import customs · Delivery order · Devanning & tally · Delivery

Money words

BAF fuel · **THC** terminal handling · **CAF** currency · **PSS** peak season · **GRI** rate rise · **D&D** demurrage & detention · **TT** bank transfer · **LC** letter of credit (UCP 600) · **D/P**, **D/A** · **Landed cost**

Sri Lanka at a glance

- Declarations: **CUSDEC** in **ASYCUDA World**, by a **CHA**
- Sea manifests: **ASYHUB** only, since 7 May 2025; 72 h before arrival
- VAT 18% (from 1 Jan 2024); SSCL generally 2.5%
- Tariff: Sri Lanka Customs National Imports Tariff Guide

On arrival, always…

1. Check the seal number matches the BL
2. Photograph before opening
3. Count and scan every package
4. Weigh; flag anything over tolerance
5. Write problems on the receipt
6. Notify carrier and insurer in writing, fast
7. Return the empty before free time ends

Remember!

- Look up any word here or in Ternary Cargo's **Glossary** page.
- Keep the cheat sheets handy: containers, Incoterms, documents, money words, Sri Lanka rules.
- Rates, tariffs and rules change. Check the latest official source before a real shipment.

SOURCES

Where the facts come from

The key numbers and rules in this booklet were checked against these sources on 31 October 2026.

Topic	Sources
Containers (dry, high cube, reefer, open top, flat rack)	Hapag-Lloyd container specifications: hapag-lloyd.com/en/services-information/cargo-fleet/container/ (20′ Standard, 40′ Standard, 40′ High Cube, 20′ Reefer, 40′ Reefer High Cube, 20′ Open Top, 40′ Flatrack) and the Hapag-Lloyd Container Specification PDF (ISO 668 external dimensions); CMA CGM reefer technical specifications; DSV reefer container page
Container numbers	BIC, Container Identification Number: bic-code.org/identification-number/; ISO 6346
Air ULDs	Unilode ULD specification (unilode.com/uld-specification/); DSV LD3/AKE page; Korean Air Cargo ULD page
Incoterms® 2020	ICC, Incoterms® 2020 (iccwbo.org/business-solutions/incoterms-rules/incoterms-2020/); US International Trade Administration, Know Your Incoterms (trade.gov/know-your-incoterms); ICC Austria Incoterms 2020 rules; Munich Re Incoterms 2020 summary
VGM	IMO, Verification of the gross mass of a packed container (imo.org); IMO MSC.1/Circ.1475 guidelines; SOLAS Chapter VI Regulation 2, in force 1 July 2016
Volumetric weight	Maersk, Air cargo chargeable weight guide (2025); DHL Global Forwarding, Calculating chargeable weights
GS1 SSCC, labels and ASN	GS1 US, Serialized Shipping Container Codes; GS1 Logistic Label Guideline (ref.gs1.org/guidelines/logistic-label/); GS1 UK, Where should pallet labels be placed
Sri Lanka Customs	Sri Lanka Customs, Importing goods (customs.gov.lk/services/importing-goods/); Mandatory submission of manifests via ASYHUB from 7 May 2025; ASYHUB service page; Electronic Sea Cargo Reporting Regulations No. 1 of 2014; Sri Lanka Customs National Imports Tariff Guide 2024 (preamble: computation of import levies)
Sri Lanka VAT	Inland Revenue Department, VAT rate change notice (PN/VAT/2023-02) and VAT (Amendment) Act No. 16 of 2024: 18% from 1 January 2024
ETA prediction	Portcast predictive visibility pages; project44 blog on machine-learning ETAs
Ternary Cargo	Ternary Cargo prototype source code and v3 screenshots (31 Oct 2026); Ternary Cargo research brief and product vision (internal)

⭐ **Please note**

This booklet is a friendly guide, not legal, tax or customs advice. Container sizes vary by owner and manufacturer; duty rates, levies and procedures change; and carriers set their own free time, surcharges and cut-offs. HS codes shown are 6-digit international headings, not full Sri Lankan tariff lines. Liability time limits depend on the contract and law that apply. Check the latest official source, your carrier or your CHA before a real shipment. Company names in the examples (Ceylon Leaf Teas, Lanka Knit Apparel, Ceylon Freight Lines and the Ternary Cargo demo data) are made up. Incoterms® is a trademark of the International Chamber of Commerce.

Ternary SYSTEMS

Research. Invent. Deliver.

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